SLIIT UNI

THE KNOWLEDGE UNIVERSITY

04 – Local Area Networks

IE2004 Computer Networks

2026 July

Lesson outline

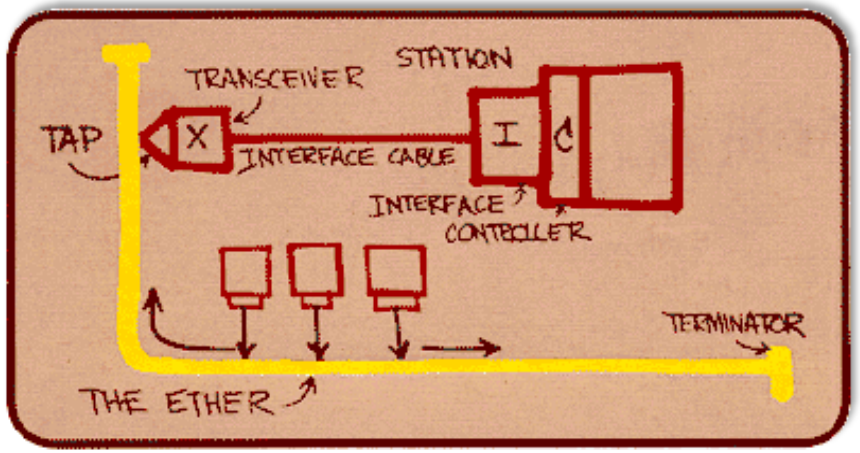
- Ethernet
- Medium access in Ethernet CSMA/CD
- Ethernet Switching
- VLANs
- Wireless LANs

Ethernet

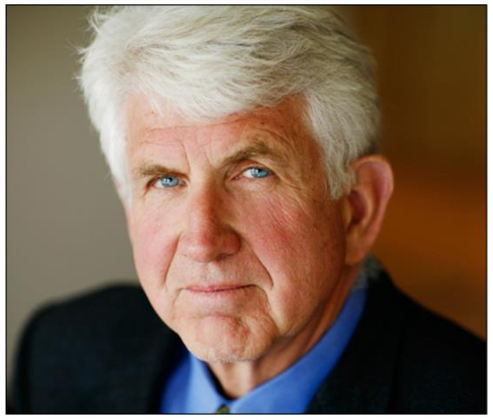
"dominant" wired LAN technology:

- first widely used LAN technology
- simpler, cheap
- kept up with speed race: 10 Mbps – 800 Gbps

Metcalfe's Ethernet sketch



Bob Metcalfe: Ethernet co-inventor, 2022 ACM Turing Award recipient

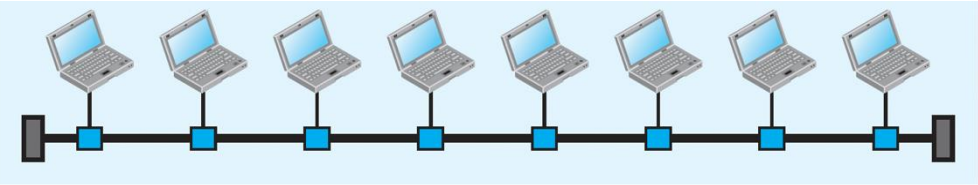


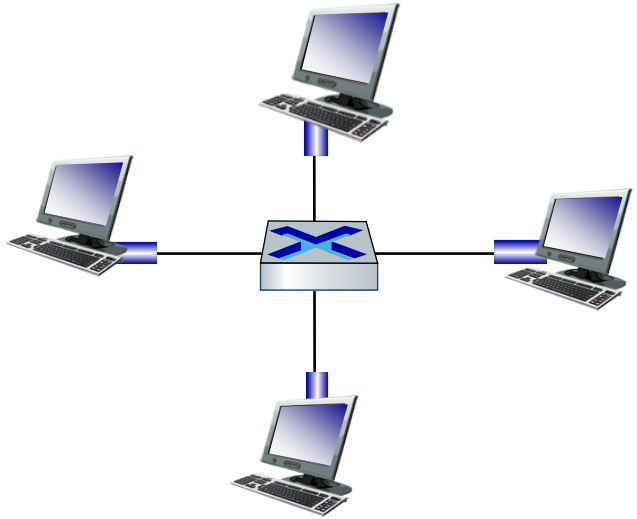
https://www.uspto.gov/learning-and-resources/journeys-innovation/audio-stories/defying-doubters

Ethernet: physical topology

- bus: from 80s to mid 90s
  - Only one host can transmit at a time
- switched: current
  - active link-layer (layer 2) *switch* in center
  - each “spoke” is now a separate Ethernet segment
  - only the switch and a host in each segment

bus: coaxial cable

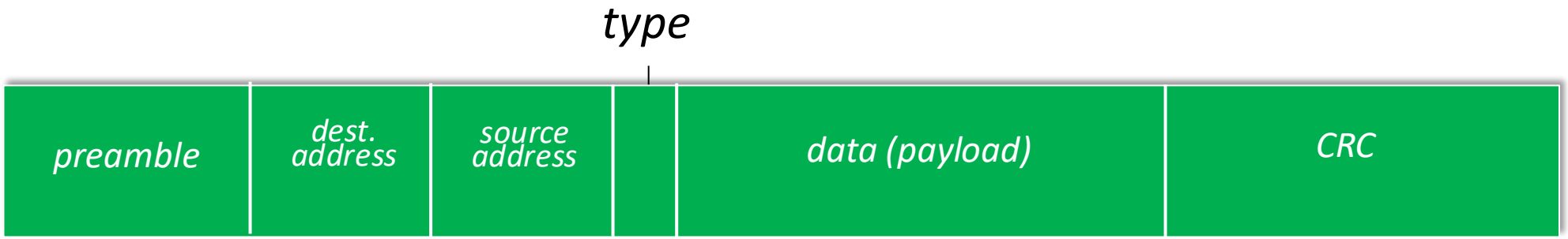




switched

Ethernet frame structure

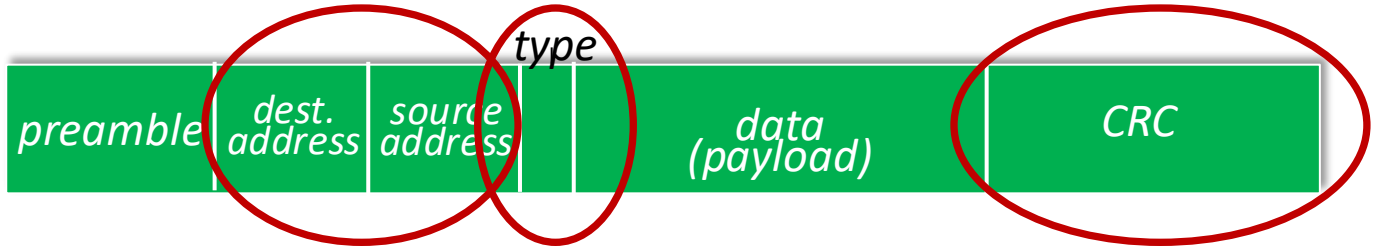
sending interface encapsulates IP datagram (or other network layer protocol packet) in Ethernet frame



preamble:

- 7 bytes of 10101010 followed by one byte of 10101011
- Used by the receiver to identify bit times

Ethernet frame structure (more)



- addresses: 6 byte source, destination MAC addresses
  - if NIC receives frame with matching destination address, or with broadcast address, it passes data (payload) in frame to network layer protocol
  - otherwise, NIC discards frame
- type: indicates network layer protocol
  - mostly IP but others possible, e.g., Novell IPX, AppleTalk
- CRC: cyclic redundancy check at receiver
  - error detected: frame is dropped

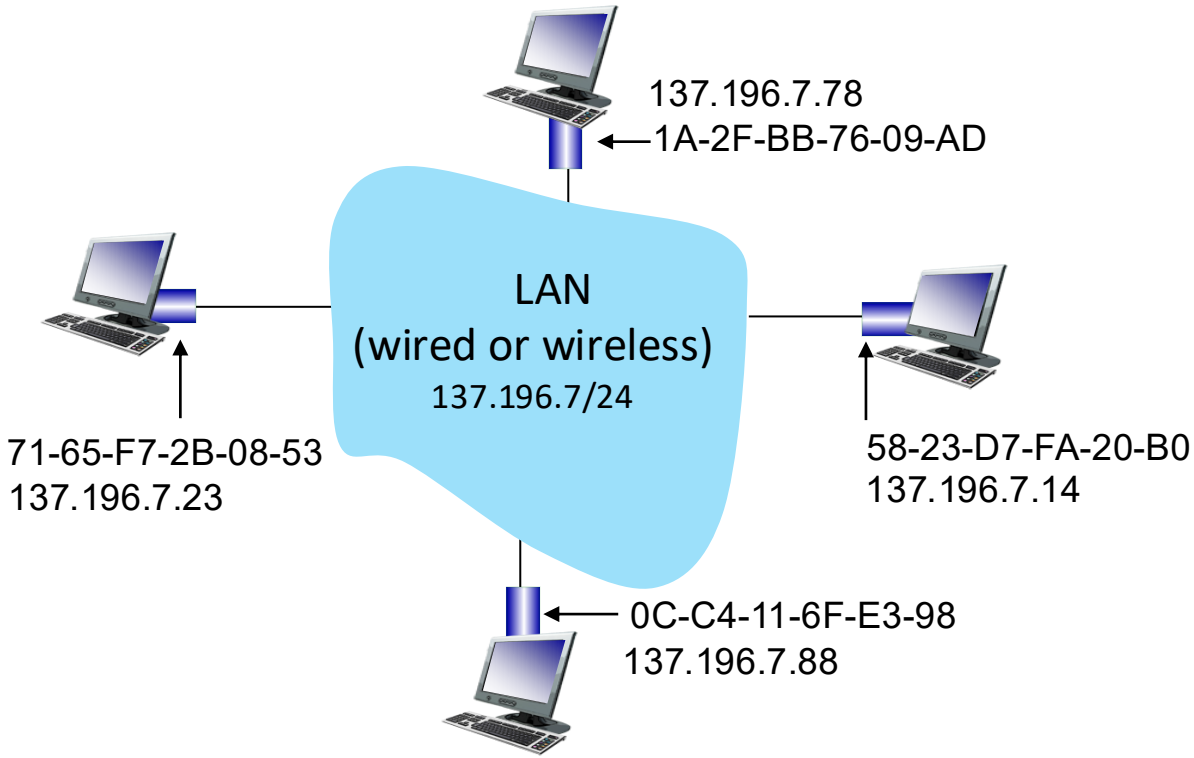
Ethernet addresses (MAC Addresses)

- Ethernet addresses are also called MAC addresses or hardware addresses
- An Ethernet address is 6 bytes long (48 bits)
- Normally expressed using hexadecimal numbers separated by colons or dashes
  - 00:80:3F:2A:17:21, 00-80-3F-2A-17-21
  - 00:80:3f:2a:17:21, 00-80-3f-2a-17-21
- Special address of all 1s is reserved for broadcast
  - i.e. FF:FF:FF:FF:FF:FF (ff:ff:ff:ff:ff:ff)
  - Can only be used as a Destination address

MAC addresses vs IP addresses

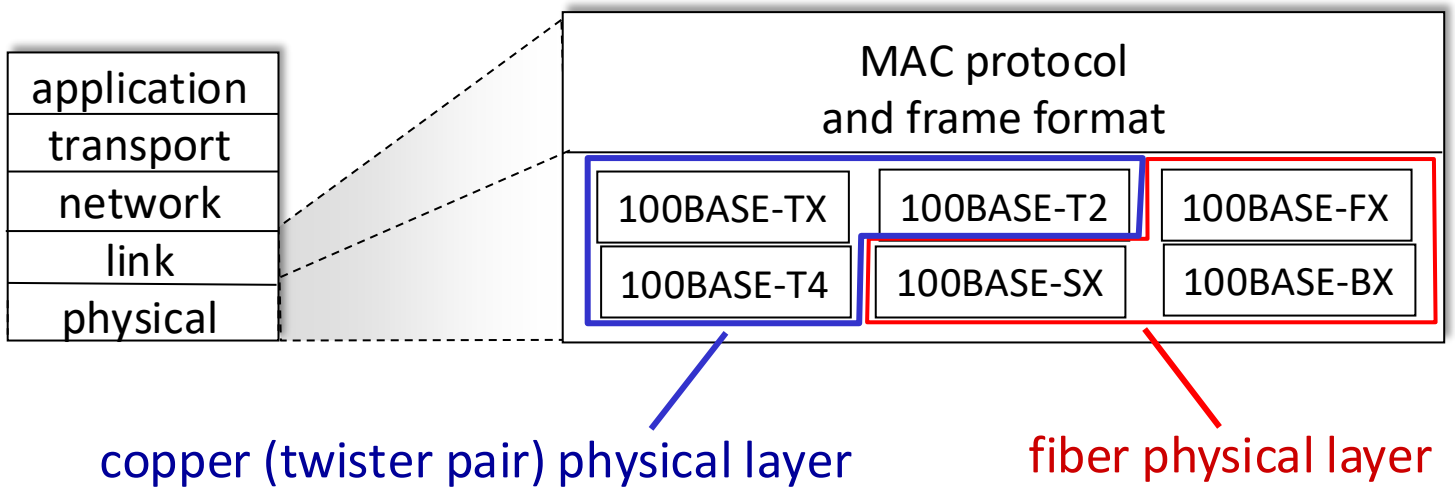
each interface on LAN

- has a unique 48-bit MAC address (layer 2 address)
  - Can be used to communicate with any device at layer 2 on the same LAN
- has a unique 32-bit IP address (layer 3 address)
  - Can be used to communicate with any device (at layer 3) on the Internet



802.3 Ethernet standards: link & physical layers

- *many* different Ethernet standards / speeds
  - common MAC protocol and frame format
  - different speeds: ..., 100Mbps, 1Gbps, 2.5Gbpbs, 5Gbps, 10Gbps, 25Gbps, 40Gbps, 50Gbps, 100Gbps, 200Gbps, 400Gbps, 800Gbps, ...
  - different physical layer media: fiber, cable.

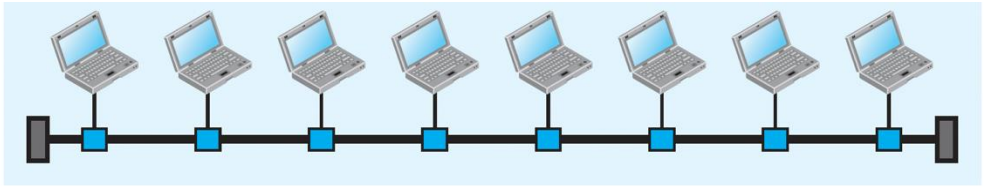


Lesson outline

- Ethernet
- Medium access in Ethernet CSMA/CD
- Ethernet Switching
- VLANs
- Wireless LANs

Ethernet medium access protocol

- Originally, Ethernet used bus topology
  - only one host could transmit at a time
  - half-duplex mode



- If more than one host transmit at the same time, a collision occurs
- Medium Access Control (MAC) protocol controls access to medium
  - when hosts are allowed to transmit
  - how to handle collisions if/when they occur
- All the hosts follow the rules in the MAC protocol
- Ethernet MAC protocol is CSMA/CD with binary exponential backoff

CSMA (carrier sense multiple access)

simple CSMA:

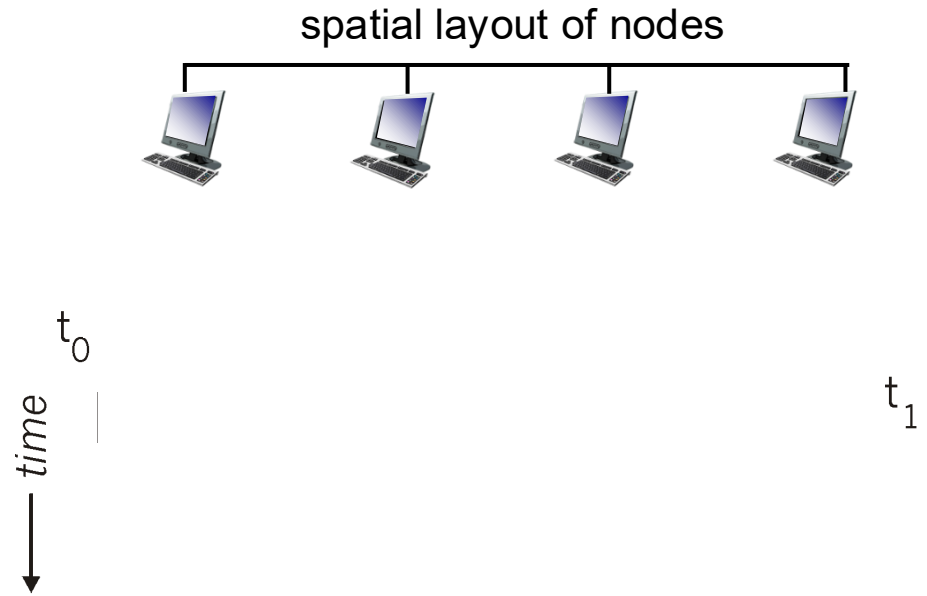
- sense the channel before transmitting
- if channel sensed idle: transmit entire frame
- if channel sensed busy: defer transmission
- human analogy: speak only when no one else is speaking!

CSMA/CD: CSMA with *collision detection*

- hosts sense channel while transmitting
- collision detection: if signal on the channel is different from own transmitted signal, there must be a collision
- colliding transmissions aborted, reducing channel wastage
- human analogy: if two persons start speaking at the same time, both with stop and try again

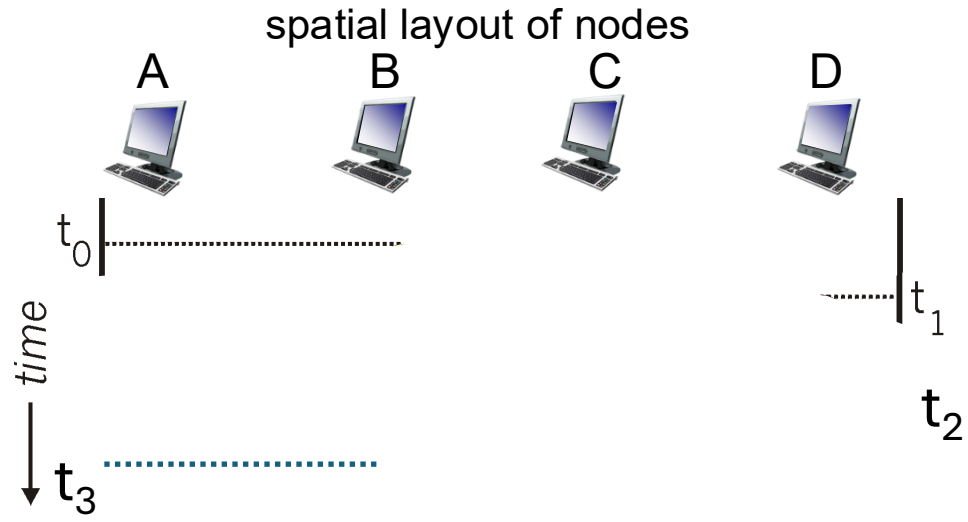
CSMA: collisions

- collisions can *still* occur with carrier sensing:
  - propagation delay means two nodes may not hear each other's just-started transmission
- collision: entire frame transmission time wasted
  - distance & propagation delay will determine the probability of a collision



CSMA/CD:

- CSMA/CD reduces the amount of time wasted in collisions
  - transmission aborted on collision detection
  - D detects collision at t_2 and aborts its transmission
  - B detects collision at t_3 and aborts its transmission



Ethernet CSMA/CD algorithm

1. When a datagram is received from network layer, create a frame
2. Sense the channel:
   if idle: start frame transmission
   if busy: wait until channel idle, then transmit
3. If entire frame transmitted without collision - done!
4. If another transmission detected while sending: abort, send jam signal
5. After aborting, enter *binary (exponential) backoff:*
   - after m^{th} collision, choose K at random from $\{0,1,2, \ldots, 2^m-1\}$ wait $K \cdot 512$ bit times, return to step 2
   - more collisions: longer backoff interval

Interactive animation

Binary Exponential Backoff

- after m^{th} collision, pick a number K randomly from the range {0,1,2, …, 2^m-1}
- wait K x 512 bit times, and retry transmission
- After 1 collision a host picks K from range {0,1}
- After 2 collisions a host picks K from range {0,1,2,3}
- After 10 or more collisions a host picks K from range {0,1,…,1023}
- If two hosts pick the same number, another collision is likely
- But the range increases with the number of collisions (m)
- So, the chance of two hosts picking the same number gets lower as the number of collisions (m) increases

Review questions

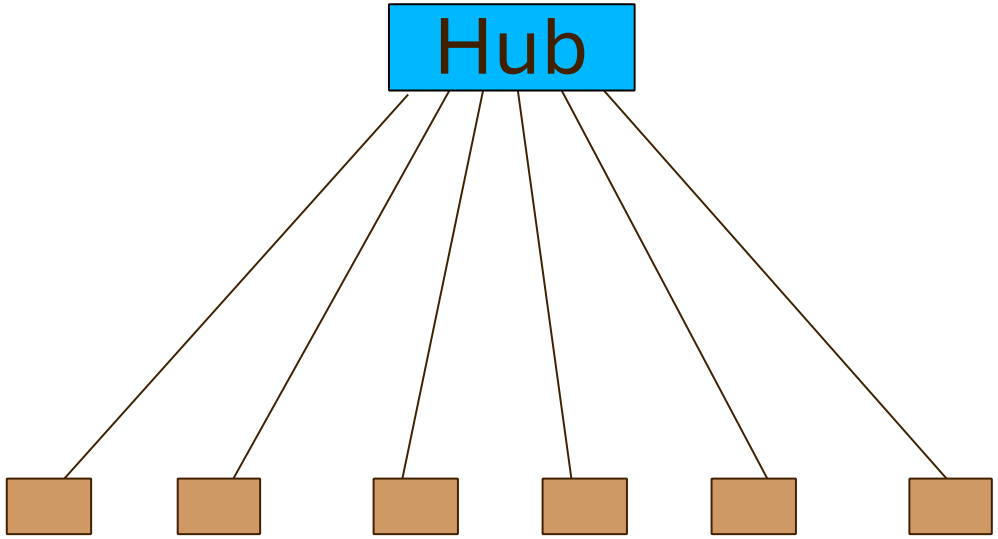
- Describe the Ethernet frame format.
- What is the format of an Ethernet address?
- Does Ethernet support broadcast and multicast?
- How is collision detection achieved in CSMA/CD?

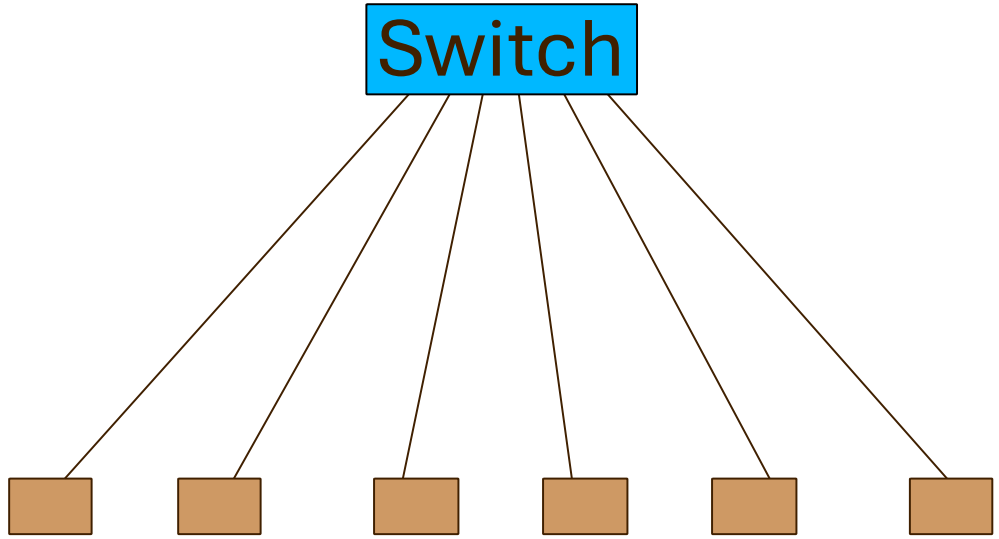
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Ethernet networks using star topology

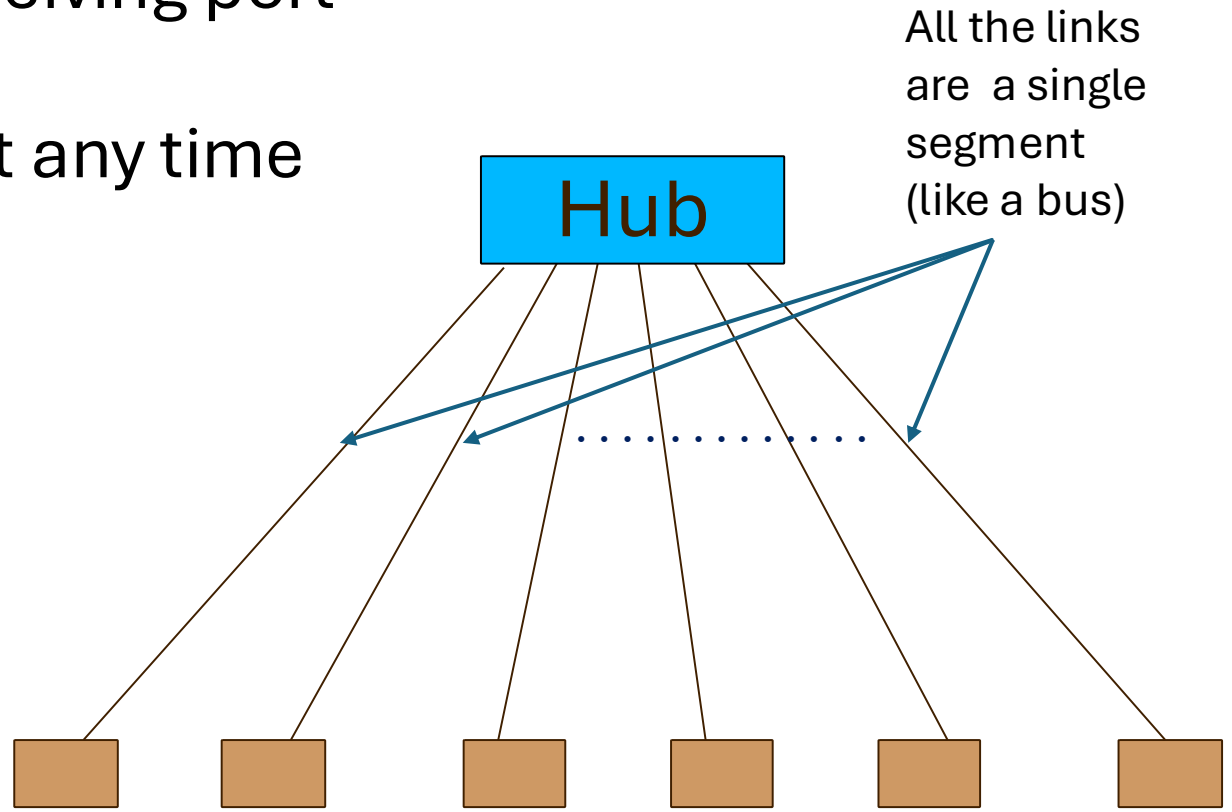
- Due to many problems in bus topology and coaxial cable, topology of Ethernet networks changed to star topology in the 90s
- Initially used hubs to interconnect hosts
- Later used switches instead of hubs
- Interconnection done using UTP cable





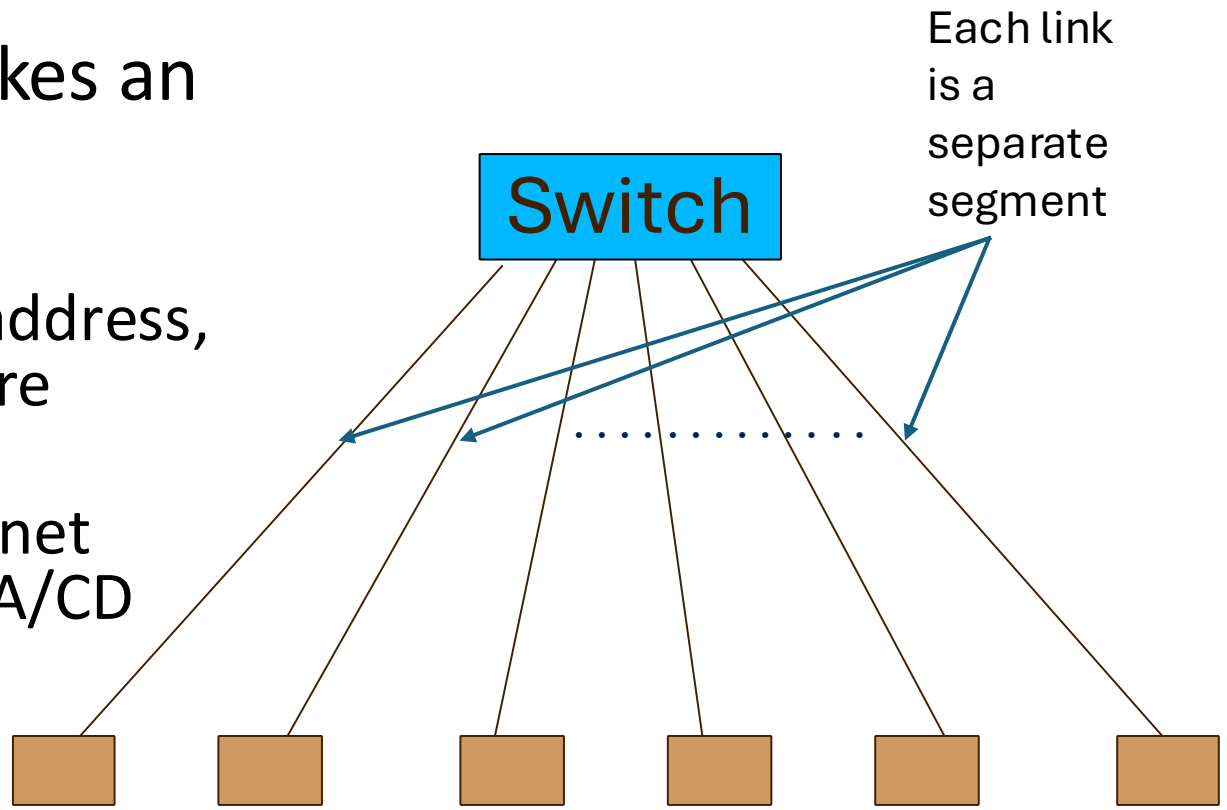
Ethernet hubs / collision domains

- Ethernet hubs are physical layer devices
- When a frame is received, it copies the signal to all the ports except the receiving port
- Only one of the devices in the entire network can transmit at any time
- If two devices transmit at the same time, a collision occurs
- All devices have to operate in half-duplex mode
- The entire network is a single collision domain



Ethernet switches

- Switch is a link-layer device: takes an *active* role
  - store, forward Ethernet frames
  - examine incoming frame's MAC address, and forward frame to one-or-more outgoing links
  - Each segment is a separate Ethernet network following Ethernet CSMA/CD protocol, independent of other segments
- plug-and-play, self-learning
  - switch and the attached host can do auto negotioation

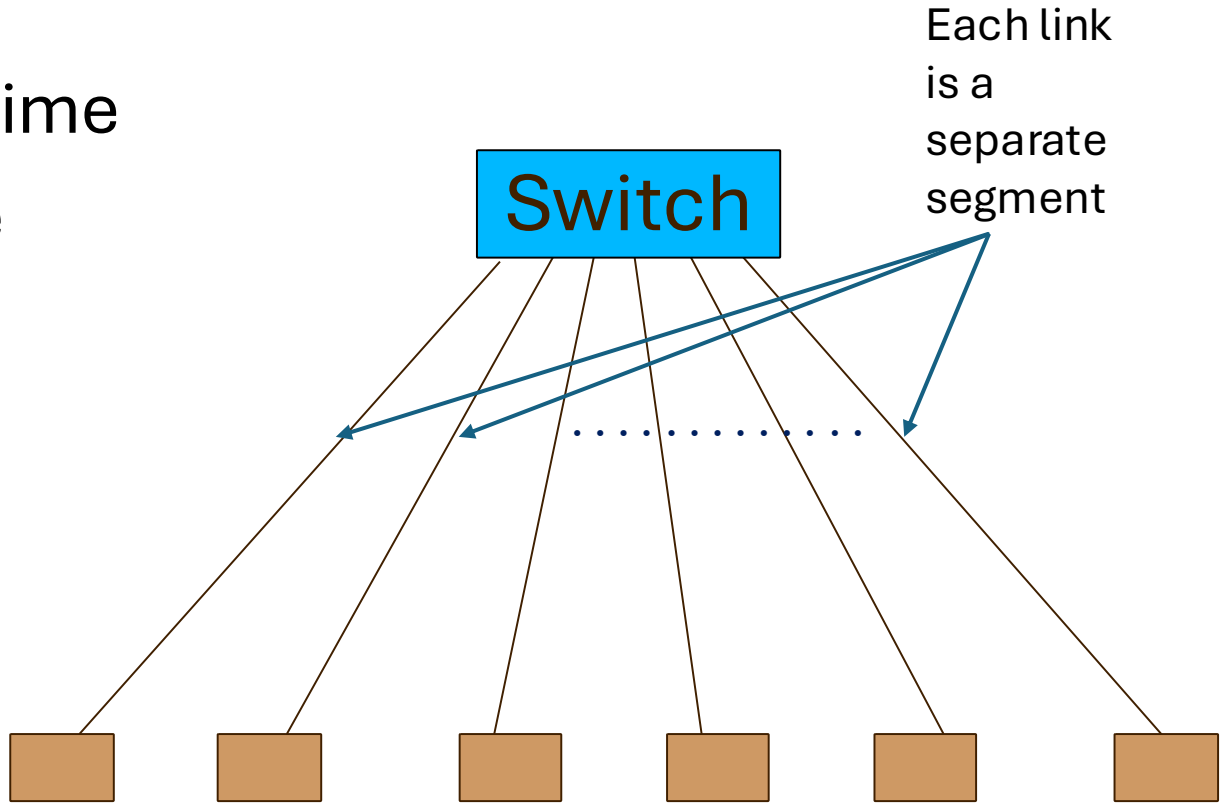


Half-duplex mode / collision domains

- Ethernet CSMA/CD rules applicable to each segment separately

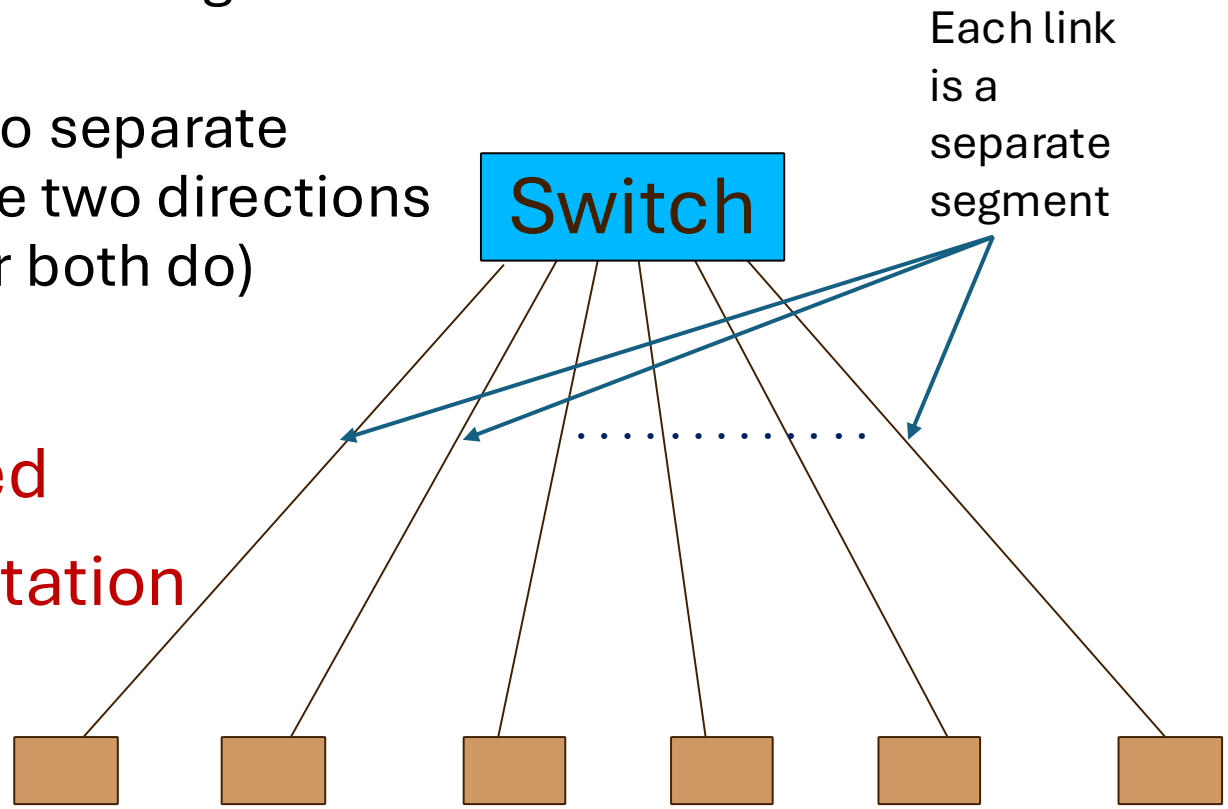
If a segment operates in half-duplex mode:

- Only one of the devices in the segment can transmit at any time
- If both devices transmit at the same time, a collision occurs
- Each segment is a separate collision domain



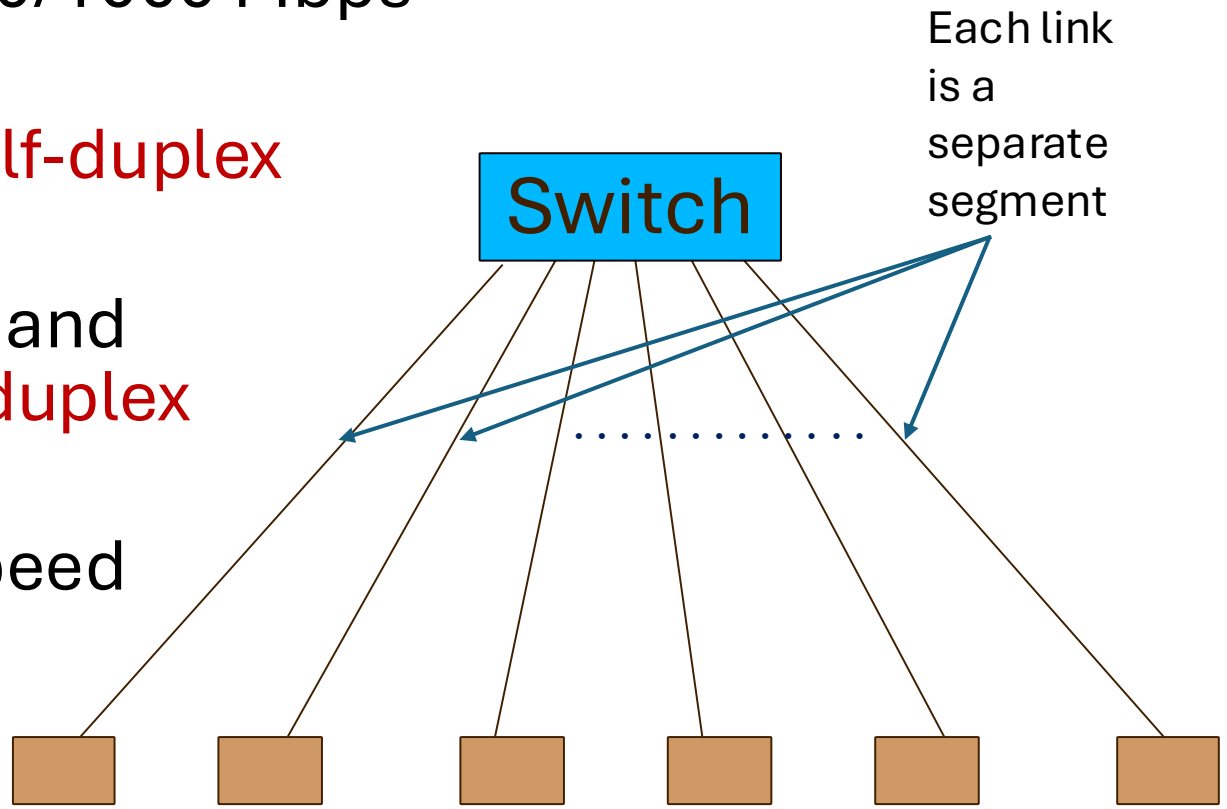
Full duplex mode

- Full duplex operation is possible when
  - both devices in a segment are capable of full duplex mode (i.e. sending and receiving at the same time)
  - the physical media provides two separate channels for sending data in the two directions at the same time (UTP and fiber both do)
- No collisions
- CSMA/CD protocol not needed
- Both the switch and the end station can transmit at any time



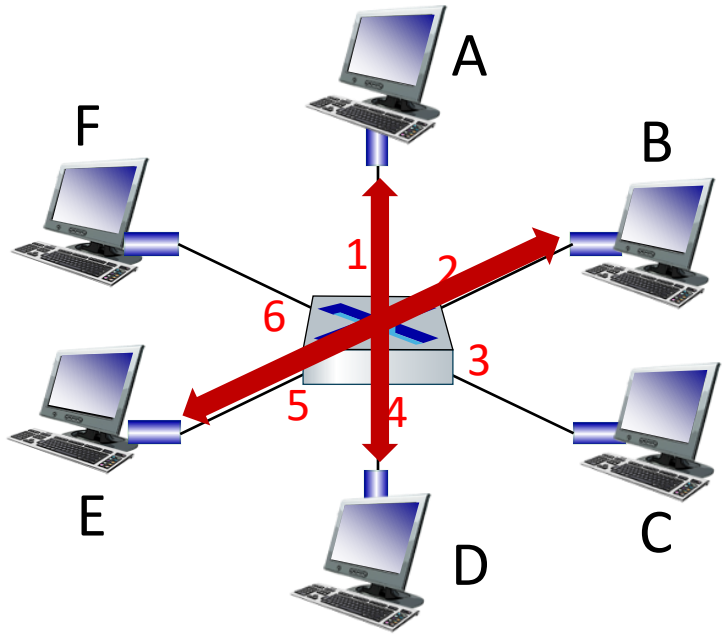
Ethernet auto-negotiation

- Only two devices in each segment
  - Host (PC, notebook, phone etc) and the switch
- Modern switches support 10/100/1000 Mbps speeds for all ports
- Each segment can operate in half-duplex mode or full-duplex mode
- When a host connects, the host and the switch will perform speed / duplex negotiation
- They will agree on the highest speed and duplex mode that both of them can operate at



Switch: multiple simultaneous transmissions

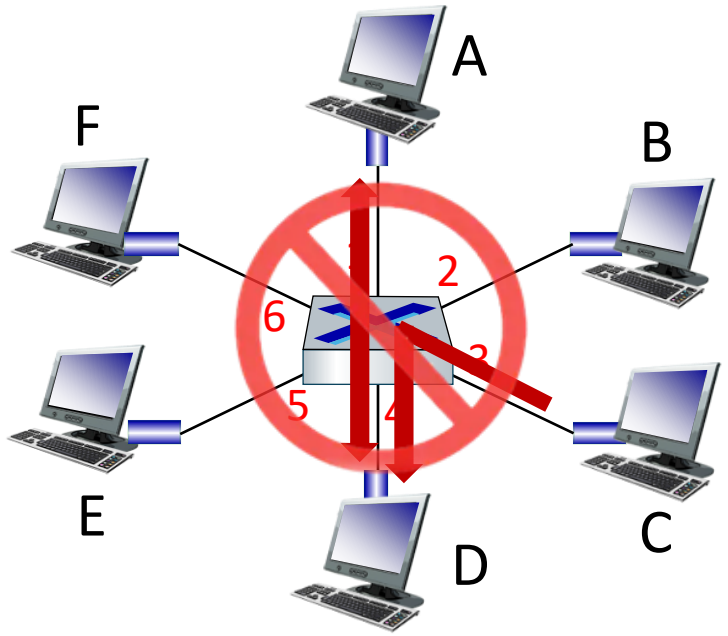
- hosts have dedicated, direct connection to switch
- switches buffer frames
- Ethernet protocol used on *each* incoming link, so:
  - each link is its own collision domain
  - no collisions if full duplex
- switching: A-to-D and B-to-E can happen simultaneously, without collisions



switch with six ports (1,2,3,4,5,6)

Switch: multiple simultaneous transmissions

- switching: A-to-D and C to D can *not* happen simultaneously
- Switch cannot transmit two frames in link 4 at the same time
- One of the frames will be buffered by the switch
- Send the buffered frame only after the first frame has been sent



switch with six ports (1,2,3,4,5,6)

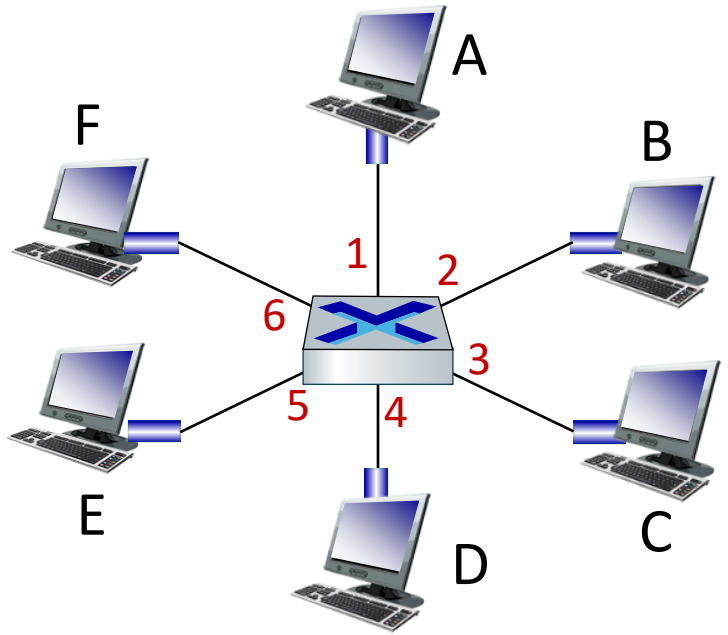
Switch MAC address table

Q: how does switch know D is reachable via port 4, E is reachable via port 5?

A: each switch has a MAC address table with three parts in each entry:

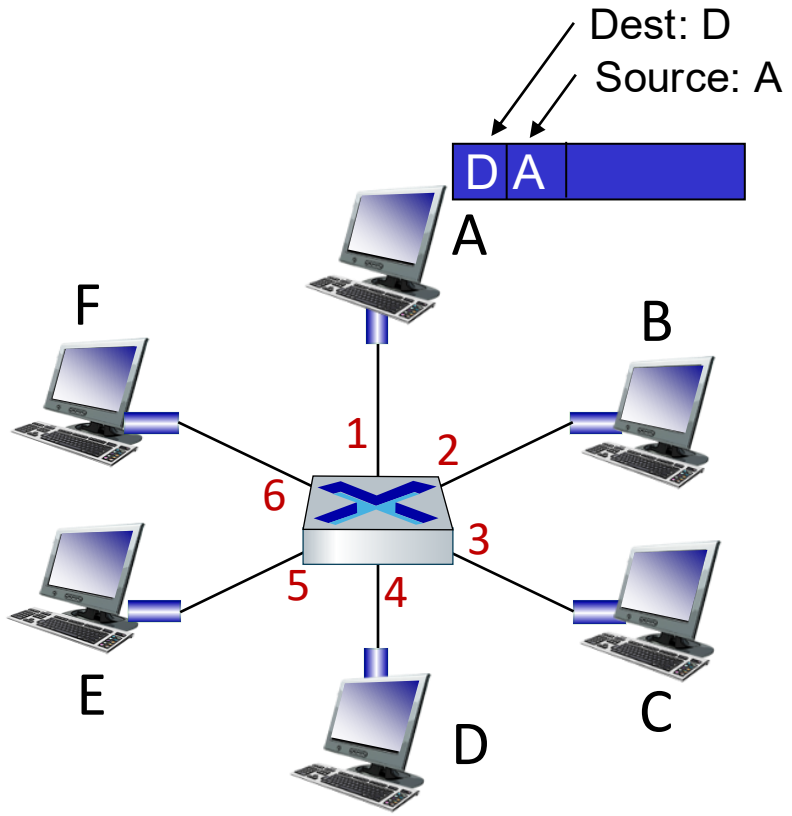
- MAC address of host, port to reach host, time stamp

Q: how are MAC address table entries created and maintained?



Switch: self-learning

- switch *learns* which hosts can be reached through which ports
- when frame received, switch "learns" location of sender: incoming port number
- records sender MAC address, port in table
- TTL of each new entry set to some value
- TTL field of existing entries decremented every second
- Entries get removed when TTL becomes 0



MAC Address table (initially empty)

MAC addr	port	TTL
A	*1*	*60*

Switch: address learning and forwarding

when a switch receives a frame:

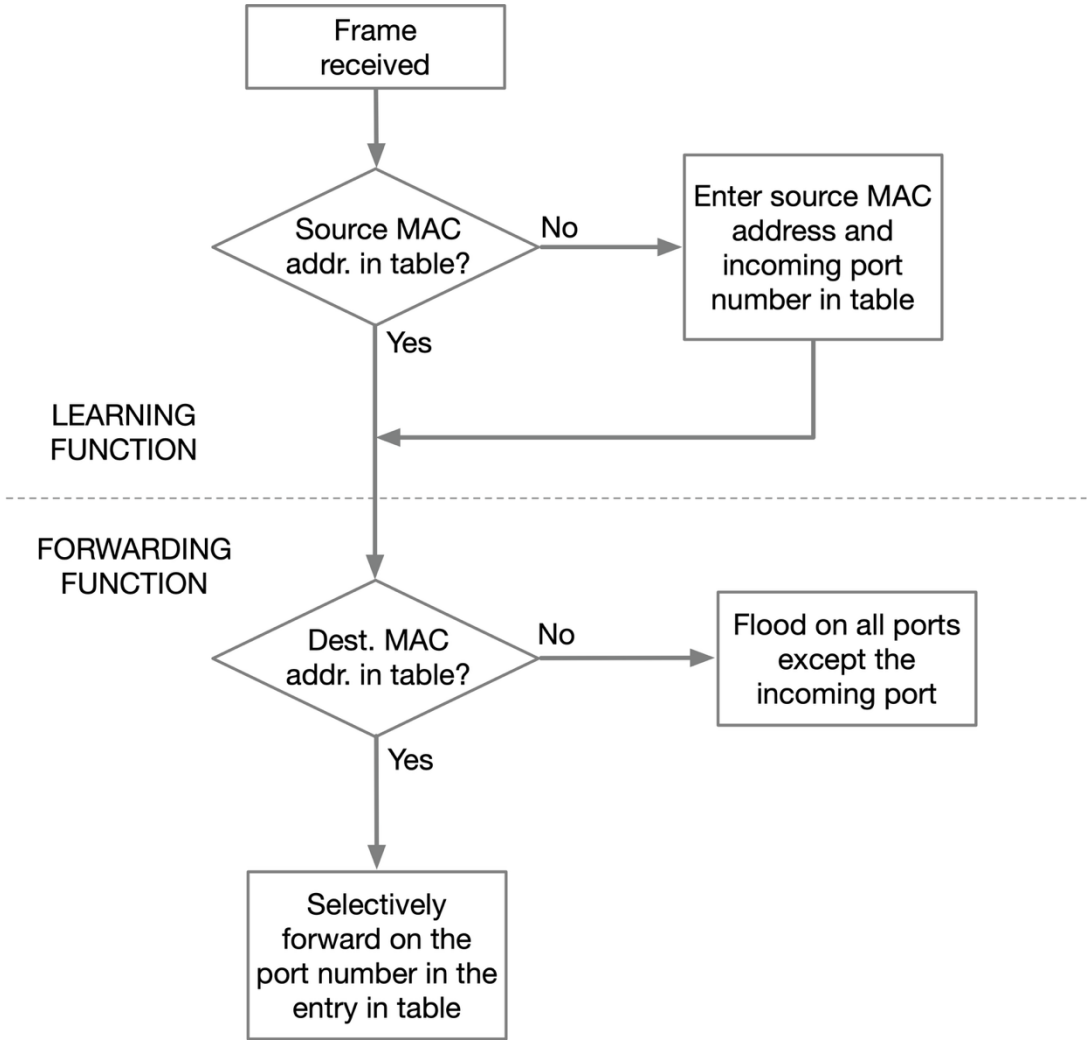
Learning

1. If Source MAC address is not in the MAC address table, add the source MAC address and incoming port ID to table
2. If MAC address is there, but port number is different, update existing entry

Forwarding

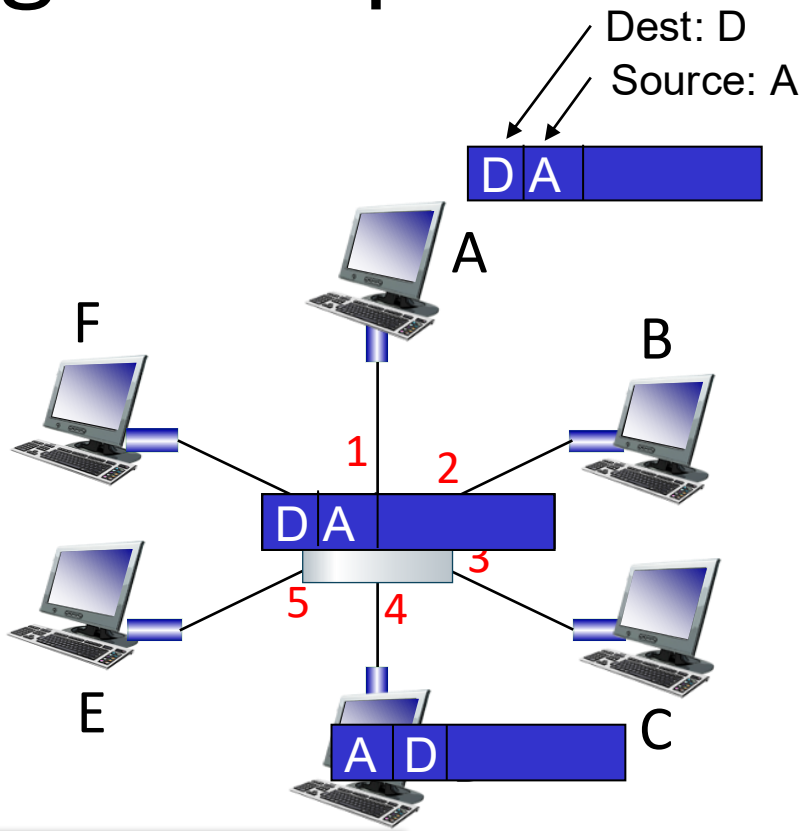
3. Search the MAC address table for the destination MAC address
4. if entry found for destination MAC address
   forward frame on port indicated by entry (selective forward)
   else
   forward on all port except arriving port (flood)
5. Broadcast frames (destination address = ff:ff:ff:ff:ff:ff) are always flooded

Address learning & Forwarding



Self-learning and forwarding: example

- Frame 1 from A to D
  - frame source A not in table: update table
  - frame destination, D,not in table: flood
- Frame 2 from D to A (5 seconds later)
  - frame source D not in table: update table
  - frame destination A in table: selectively send via port 1

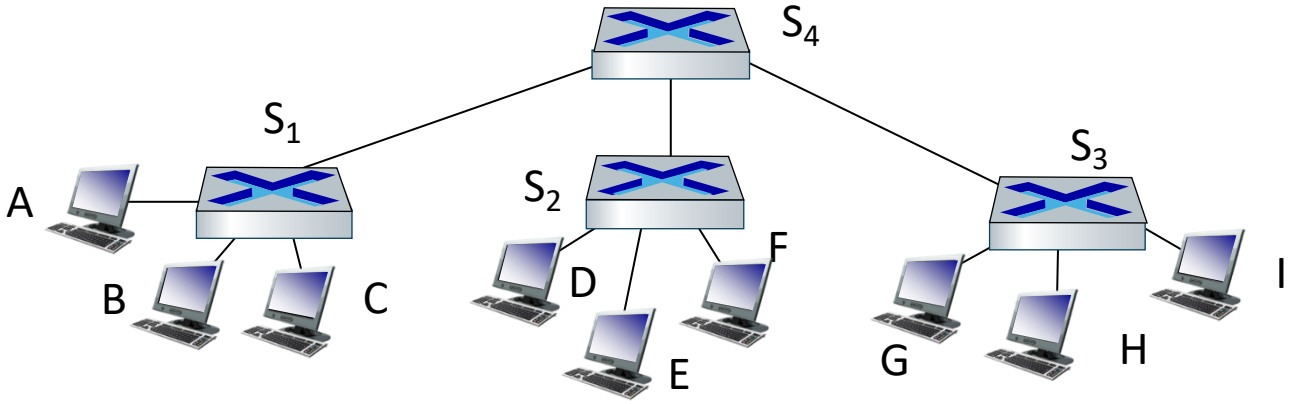


MAC addr	port	TTL
A	*1*	*55*
D	*4*	*60*

switch table (initially empty)

Interconnecting switches

self-learning switches can be connected together:

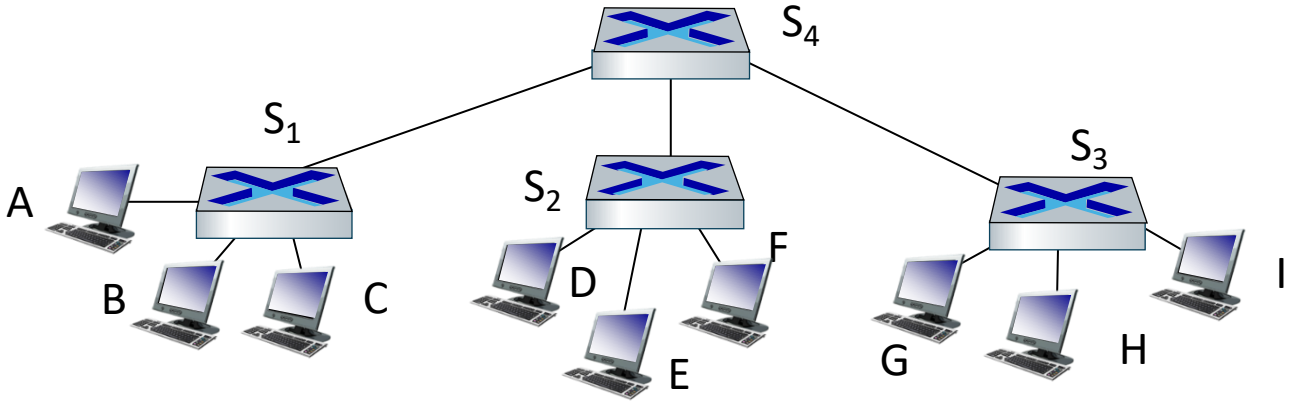


Q: sending from A to G - how does S_1 know to forward frame destined to G via S_4 and S_3?

- *A:* self learning! (works exactly the same as in single-switch case!)

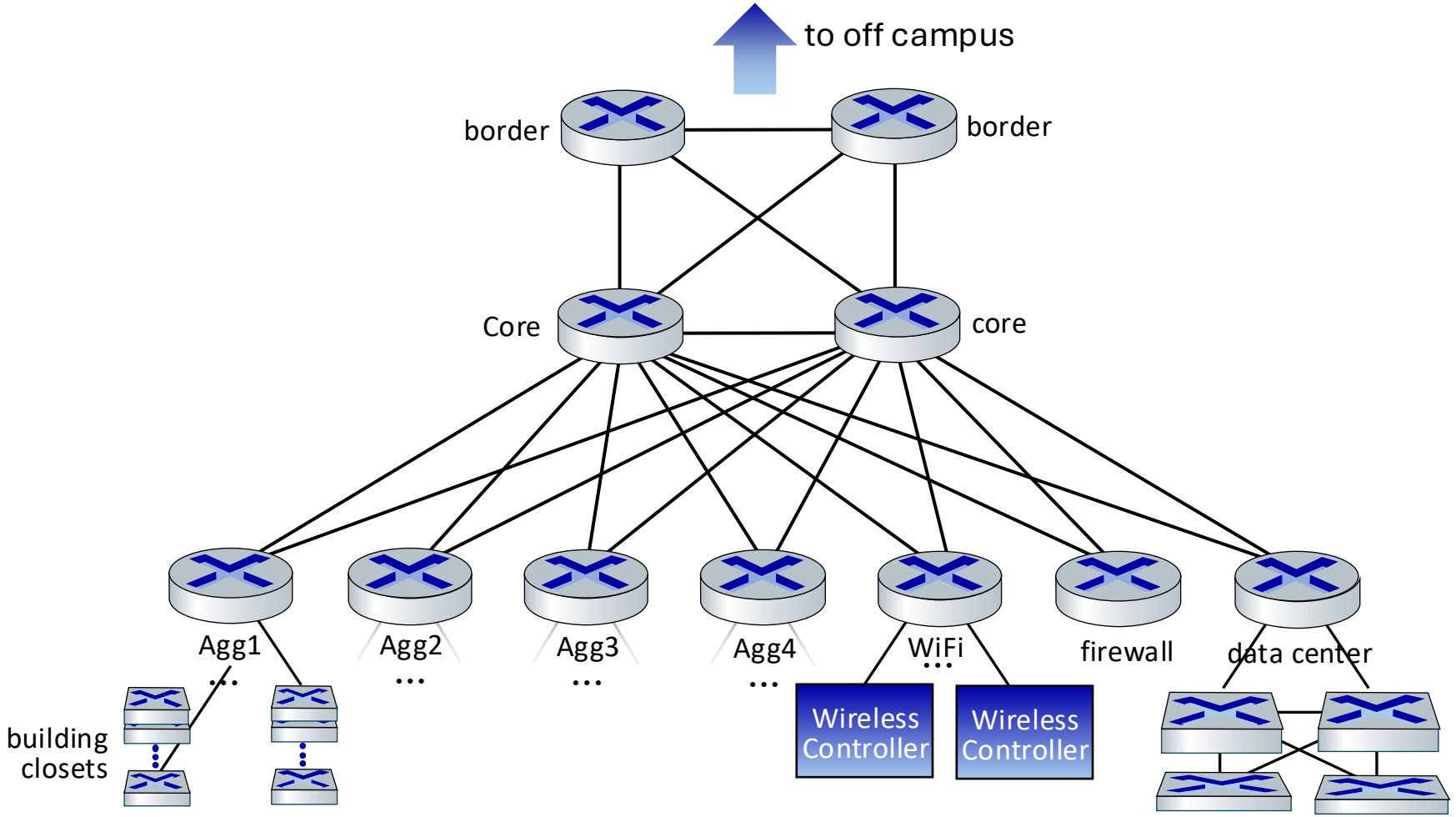
Self-learning multi-switch example

Suppose C sends frame to I and I responds to C



Q: Assume all switch tables are empty. Show how the switch tables of S_1, S_2, S_3, S_4 change when above frames are received, and show how the two frames are forwarded by S_1, S_2, S_3, S_4 (i.e. flooding or selective forwarding)

A Campus Network



Campus network:

- Interconnected LANs
  - Can be 10s or 100s of LANs
- Hierarchical
  - 3 Layers
  - Access, Distribution and Core
- Redundant
- Uses switches and routers

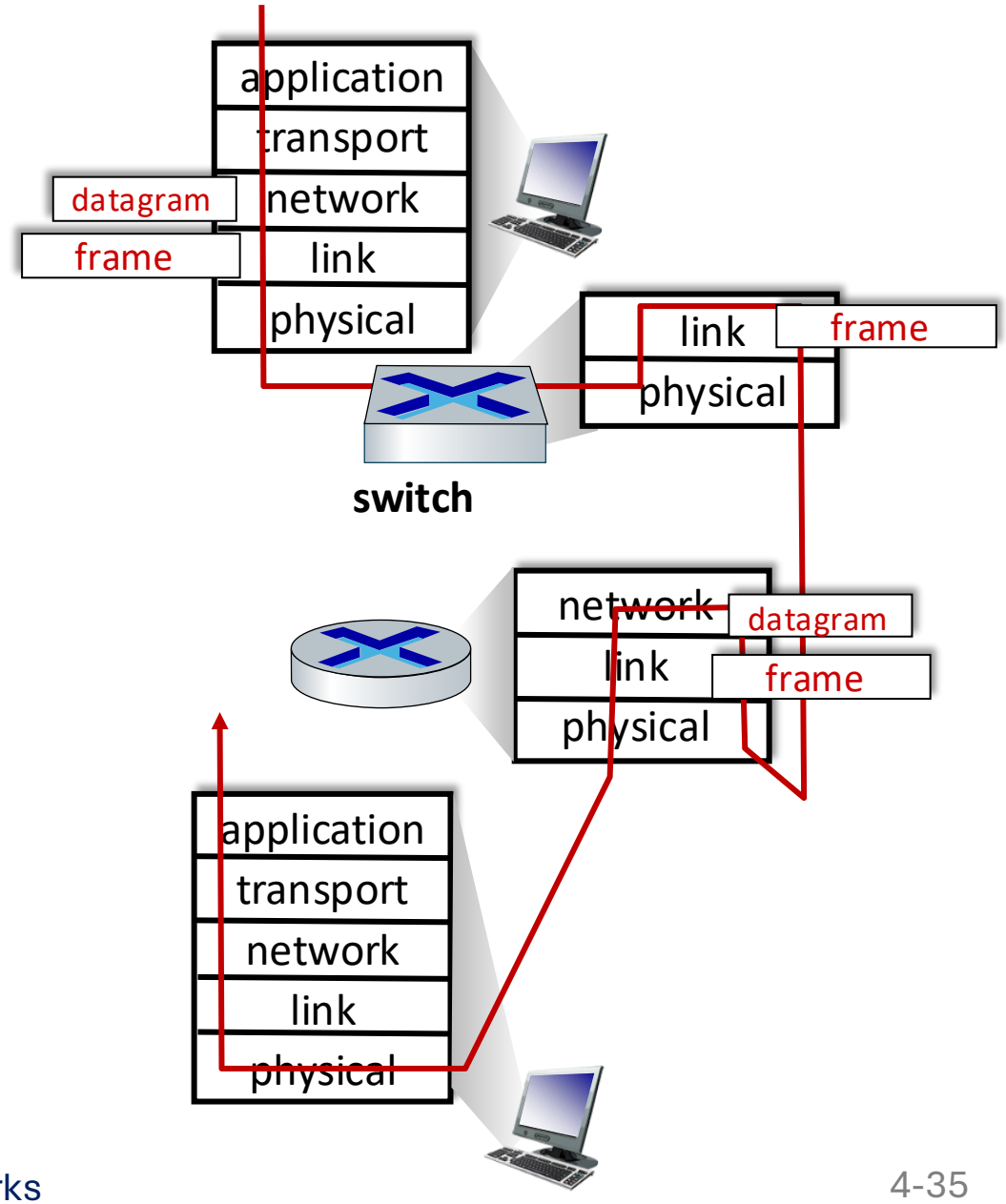
Switches vs. routers

both are store-and-forward:

- *routers*: network-layer devices (examine network-layer headers)
- *switches:* link-layer devices (examine link-layer headers)

both have forwarding tables:

- *routers:* compute forwarding tables for IP addresses using routing algorithms
- *switches:* learn forwarding table for MAC addresses using the source address of incoming frames



Review questions

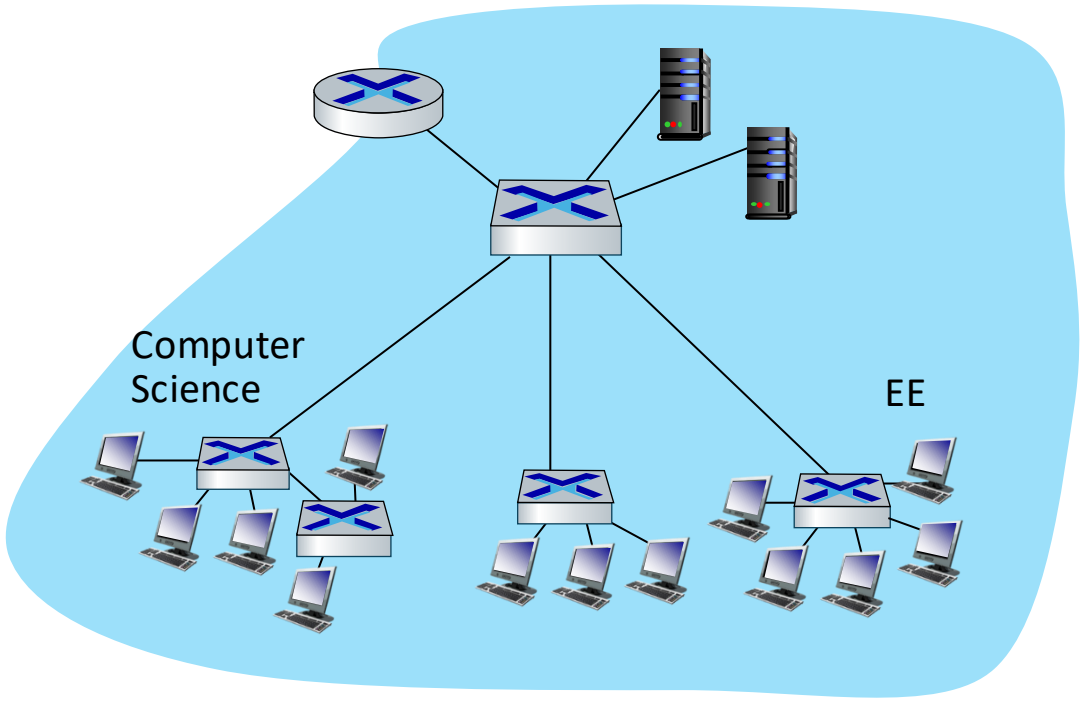
- What is Ethernet autonegotiation?
- What are the requirements that must be fulfilled for full-duplex operation in an Ethernet segment?
- How does a switch learn MAC addresses?
- How does a switch forward the following frames:
  - Unicast frame with Destination MAC not in address table
  - Unicast frame with Destination MAC in address table
  - Broadcast frame

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Virtual LANs (VLANs): motivation

Q: what happens as LAN sizes scale, users change point of attachment?

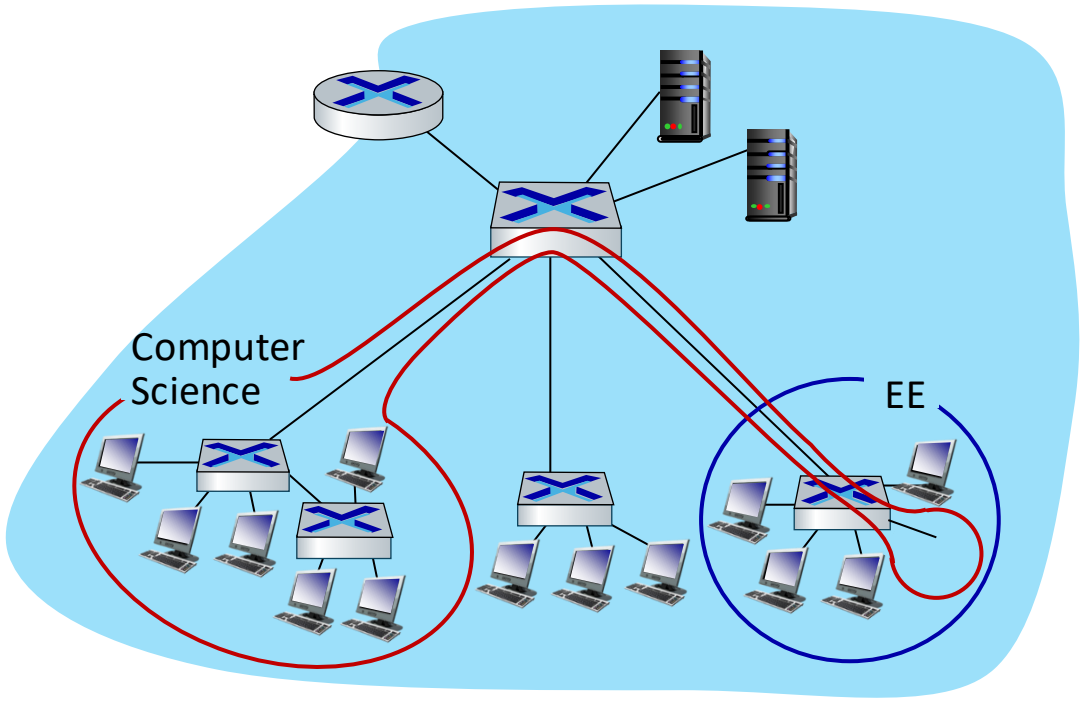


single broadcast domain:

- *scaling:* all layer-2 broadcast traffic (ARP, DHCP, unknown MAC) must cross entire LAN
- efficiency, security, privacy issues

Virtual LANs (VLANs): motivation

Q: what happens as LAN sizes scale, users change point of attachment?



single broadcast domain:

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- efficiency, security, privacy issues

administrative issues:

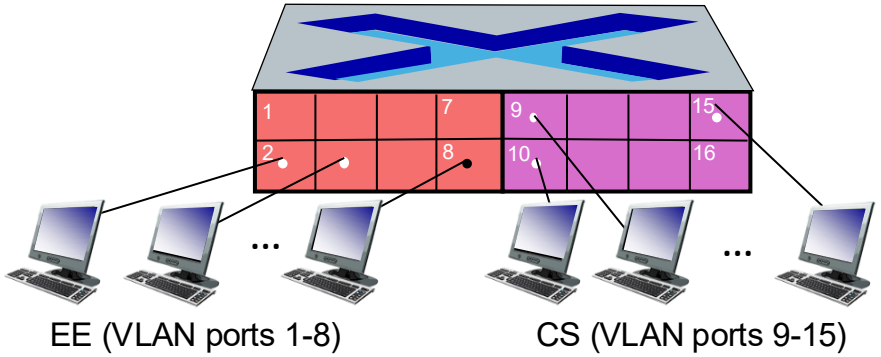
- CS user moves office to EE - *physically* attached to EE switch, but wants to remain *logically* attached to CS switch

Port-based VLANs

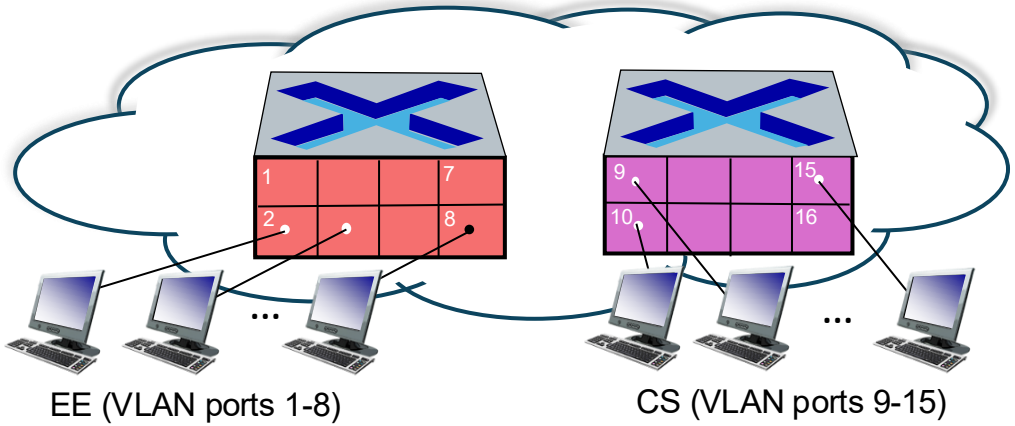
Virtual Local Area Network (VLAN)

switch(es) supporting VLAN capabilities can be configured to define multiple *virtual* LANS over single physical LAN infrastructure.

port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch ……

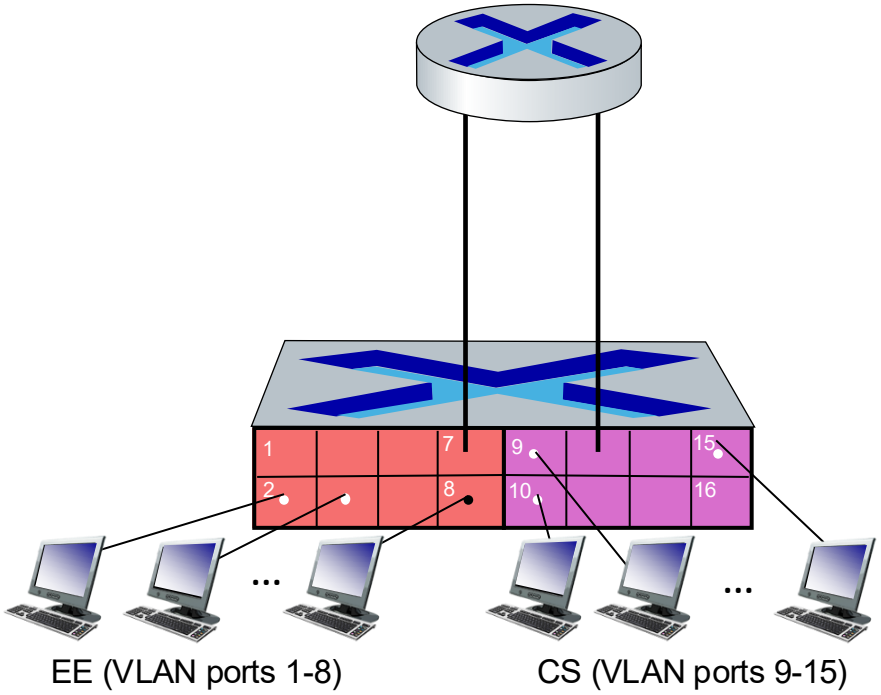


… operates as multiple virtual switches

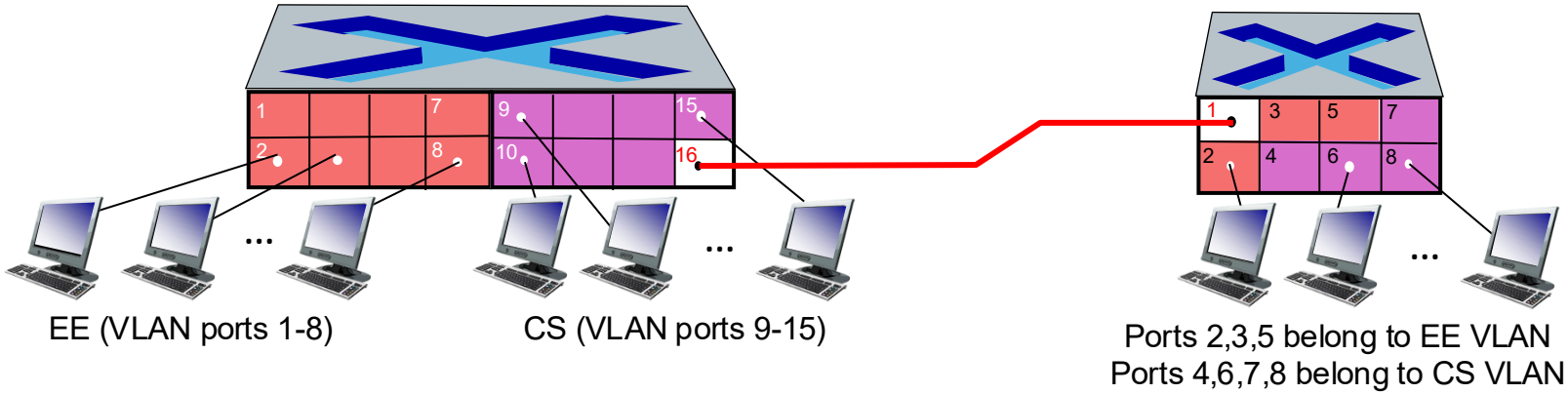


Port-based VLANs

- traffic isolation: frames to/from ports 1-8 can *only* reach ports 1-8
  - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- dynamic membership: ports can be dynamically assigned among VLANs
- forwarding between VLANS: done via routing (just as with separate switches)
  - in practice vendors sell combined switches plus routers



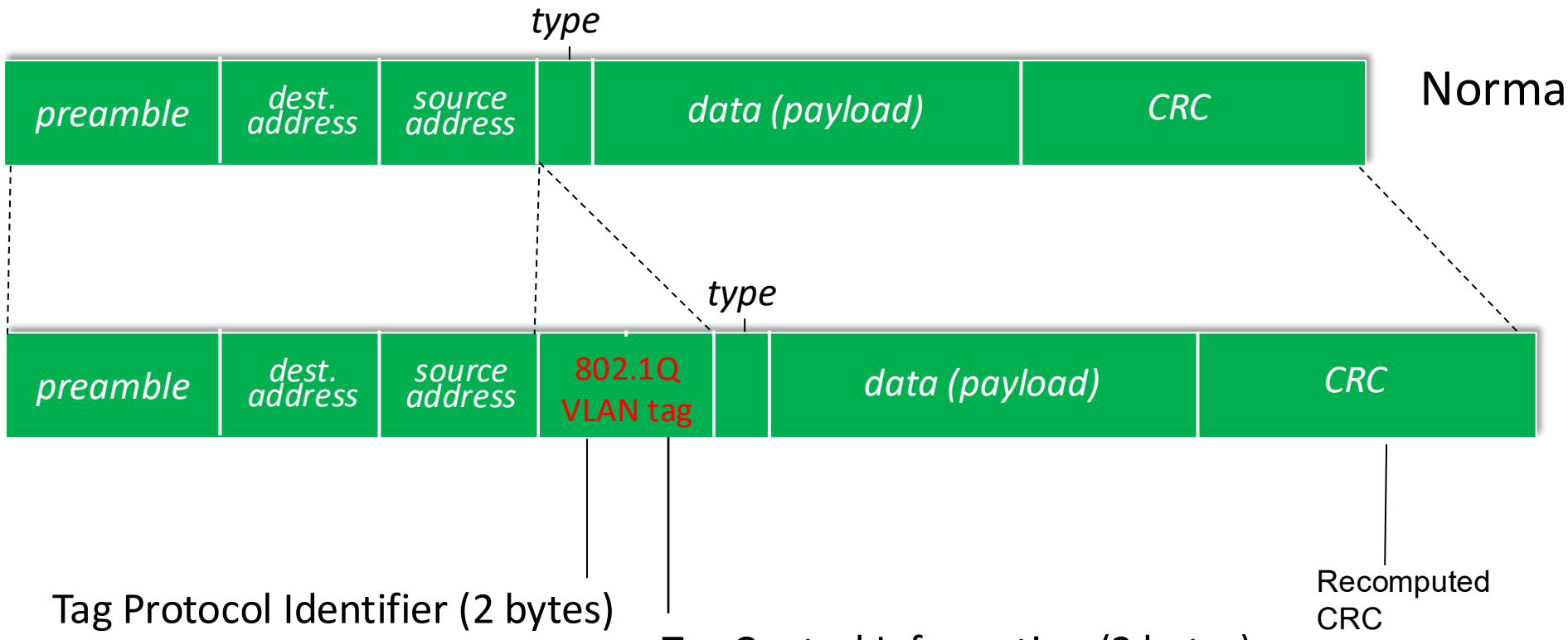
VLANS spanning multiple switches



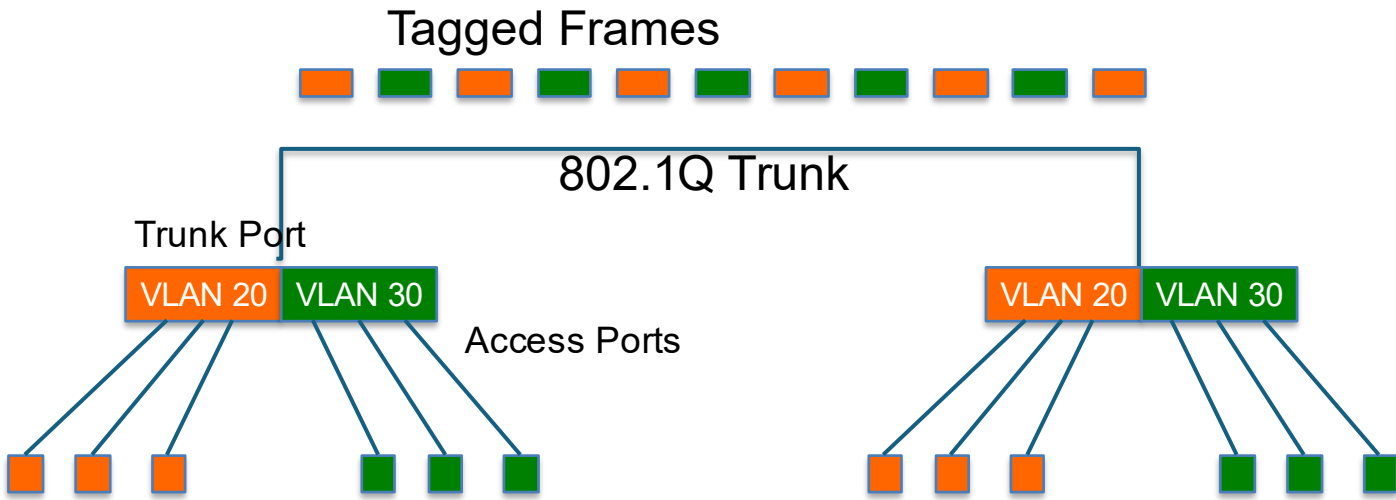
trunk port: carries frames between VLANS defined over multiple physical switches

- frames forwarded within VLAN between switches can't be normal 802.3 frames (must carry VLAN ID info)
- 802.1Q protocol adds a VLAN tag for frames forwarded between trunk ports

802.1Q VLAN Tagged Ethernet frame

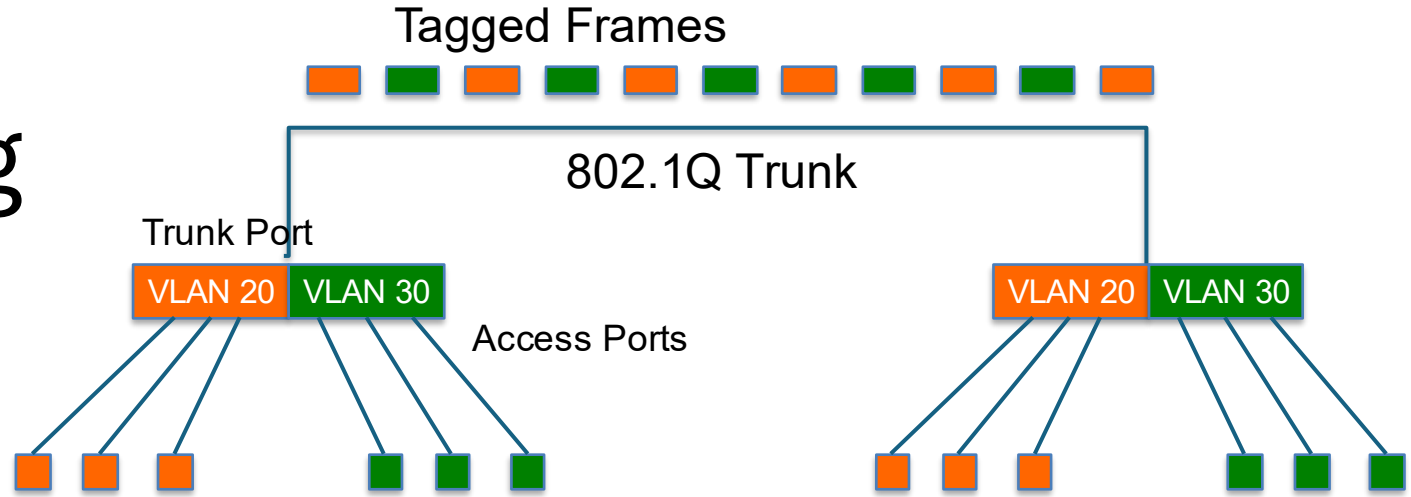


Port types



- Each port of a VLAN capable switch must belong to some VLAN
- By default, all ports of a switch belong to default VLAN (VLAN ID = 1)
- A port can be assigned to another VLAN when required
- A switch port can be an access port or a trunk port
- Access port: Switch expects to receive untagged frames and transmits outgoing frames untagged (connected to end devices)
- Trunk port: Switch expects to receive tagged frames, and transmits outgoing frames tagged (connected to switch/router)

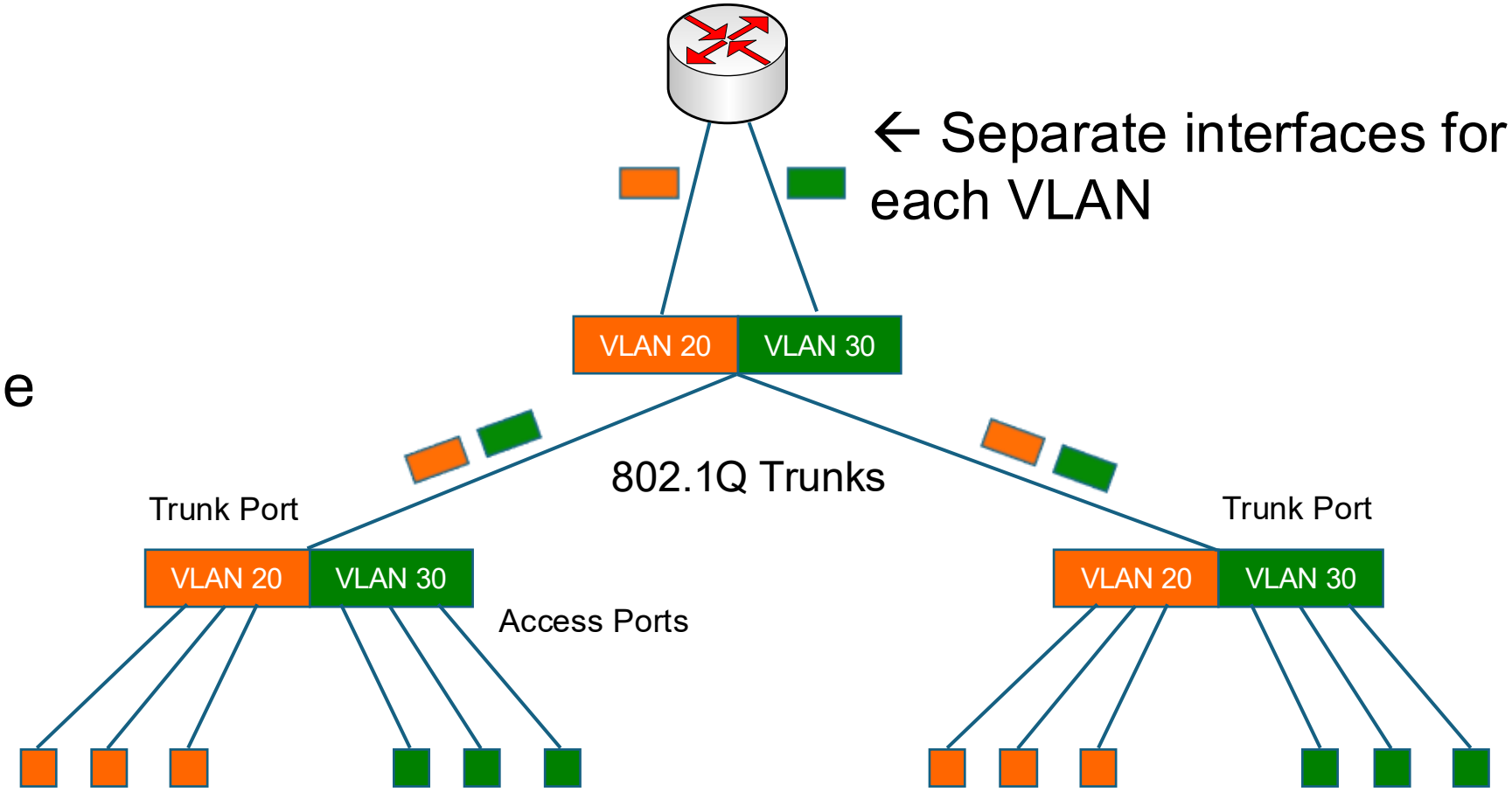
Inter VLAN Routing



- MAC address table now has VLAN number for each entry (Per VLAN MAC address tables)
- Switch will not use addresses learned in one VLAN for forwarding decisions in another VLAN
- Two hosts in two different VLANs cannot communicate with each other, although they may be on the same physical LAN
- All inter-VLAN traffic must go through a layer 3 device (router, layer 3 switch)
- Broadcast traffic in each VLAN is contained within that VLAN

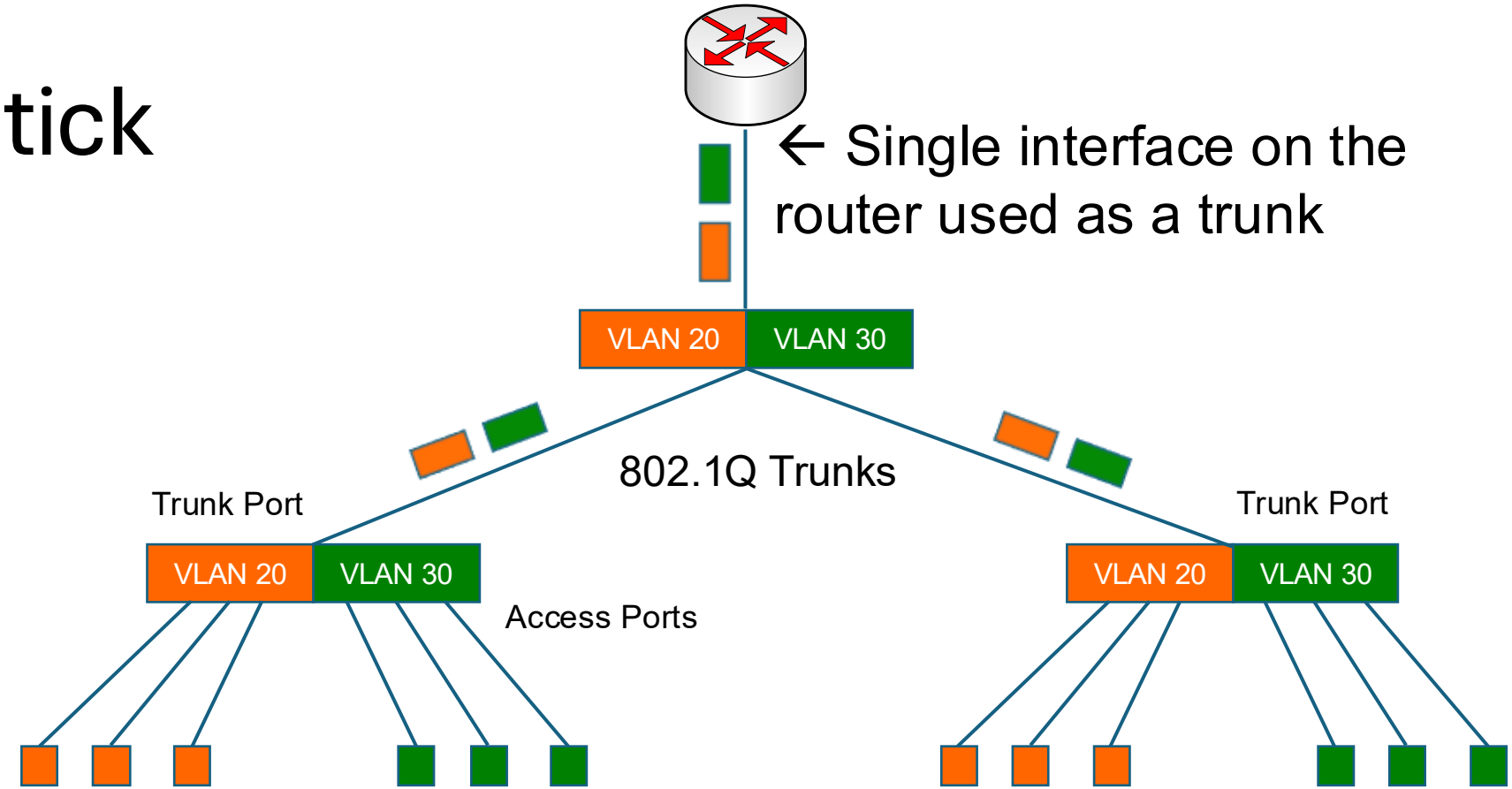
Separate interfaces for each VLAN

- Router has one interface for each VLAN
- Switch ports connected to the router are access ports



Router on a stick

- Router / L3 switch is connected to the switch via a single physical interface
- Router/L3 switch has one virtual interface for each VLAN
- Switch port connected to the router must be a trunk port



Review questions

- Where is the IEEE 802.1Q VLAN tag inserted in an Ethernet frame?
- What is an access port?
- What is a trunk port?
- What is a router on a stick?

Port types: Exercise

- Identify which ports need to be access ports and which ports need to be trunk ports
- Identify allowed VLANs in each trunk

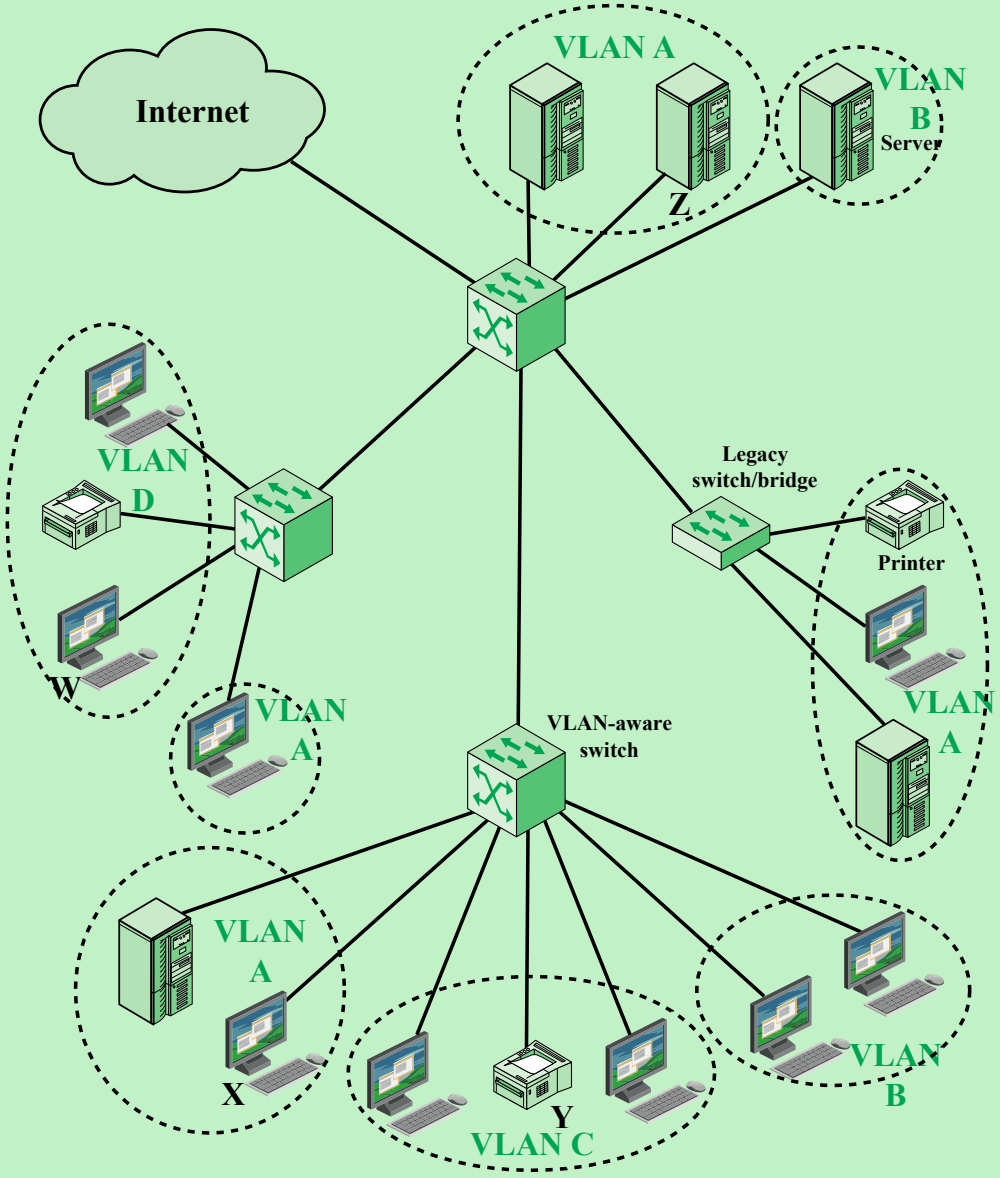


Figure 12.11 A VLAN Configuration

Lesson outline

- Ethernet
- Medium access in Ethernet CSMA/CD
- Ethernet Switching
- VLANs
- Wireless LANs
  - IEEE 802.11 standards
  - WLAN architecture
  - Medium access CSMA/CA
  - 802.11 frame format and addressing
  - Client state machine
  - Security and Authentication

IEEE 802.11 standards

- Developed by the IEEE LAN/MAN standards committee
- Covers medium access control (MAC) sublayer and physical layer
- Standards include
  - 802.11a – 1999
  - 802.11b – 1999
  - 802.11g – 2003
  - 802.11n (Wi-Fi 4) – 2009
  - 802.11ac (Wi-Fi 5) – 2013
  - 802.11ax (Wi-Fi 6) – 2021
  - 802.11be (Wi-Fi 7) – 2024
  - Several other standards
- https://standards.ieee.org/ieee/802.11/10548/

IEEE 802.11 Wireless LANs

IEEE 802.11 standard	Year	Max data rate	Range	Frequency
802.11b	1999	11 Mbps	30 m	2.4 Ghz
802.11g	2003	54 Mbps	30m	2.4 Ghz
802.11n (WiFi 4)	2009	600 Mbps	70m	2.4, 5 Ghz
802.11ac (WiFi 5)	2013	3.47Gpbs	70m	5 Ghz
802.11ax (WiFi 6)	2021	14 Gbps	70m	2.4, 5 or 6 Ghz
802.11be (WiFi 7)	2024	46 Gbps	70m	2.4, 5 or 6 Ghz
802.11af	2014	35 – 560 Mbps	1 Km	unused TV bands (54-790 MHz)
802.11ah	2017	347Mbps	1 Km	900 Mhz

- all use CSMA/CA for multiple access, and have base-station and ad-hoc network versions

Wi-Fi certification

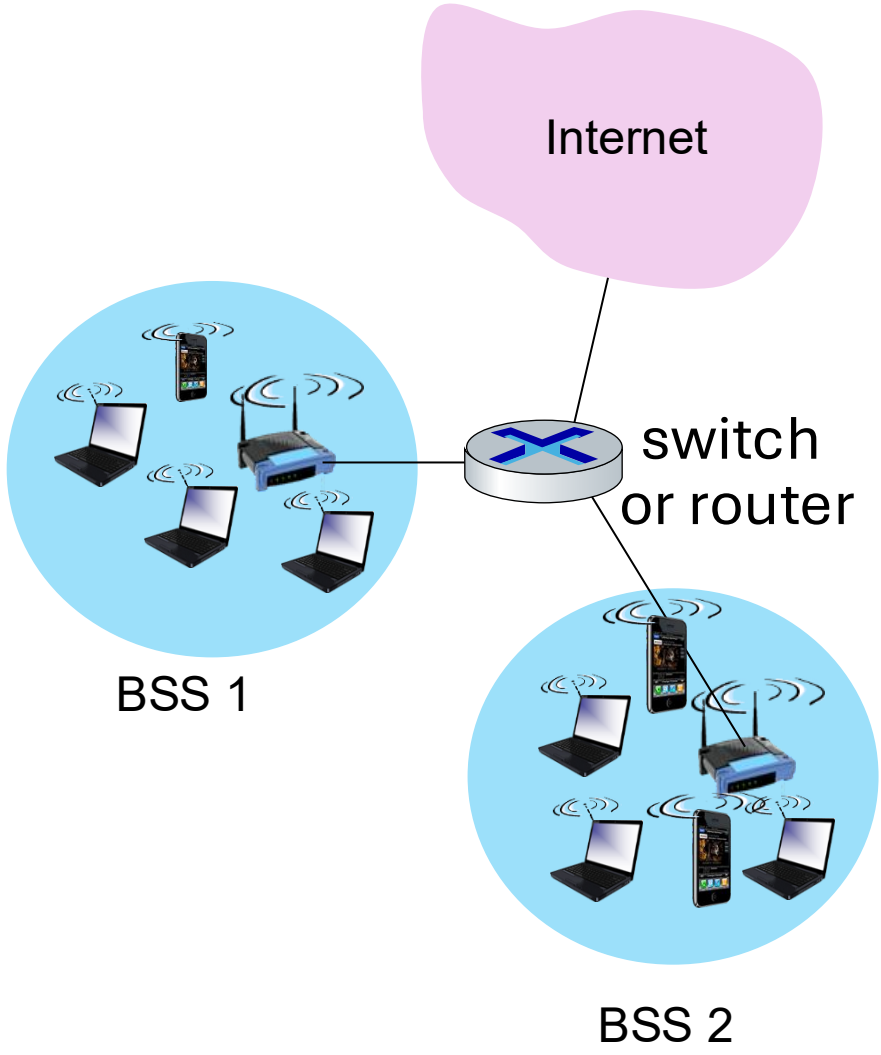


- There is always a concern whether products from different vendors will successfully interoperate
- Wireless Ethernet Compatibility Alliance (WECA)
  - Industry consortium formed in 1999
- Renamed as Wi-Fi (Wireless Fidelity) Alliance
  - Created a test suite to certify interoperability for 802.11 products
- Products that passed the test are Wi-Fi® certified
- https://www.wi-fi.com/

IEEE 802.11 architecture

- The IEEE standard defines two kinds of service sets
  - Basic service set (BSS)
  - Extended service set (ESS)
- Basic service set (BSS)
  - made up of stationary or mobile wireless stations
  - and an optional central base station, known as the access point (AP)
- Extended service set (ESS)
  - made up of two or more interconnected BSSs with APs
  - Connections between the Aps may be wired or wireless

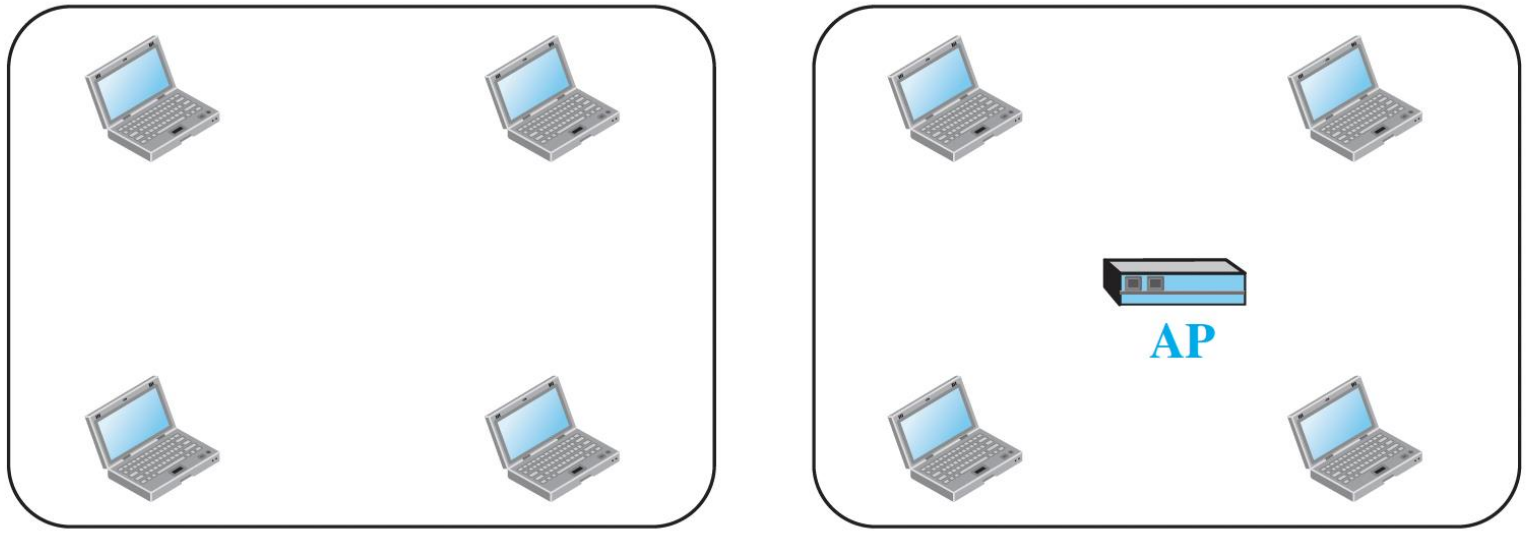
802.11 LAN architecture



- wireless host communicates with base station
  - base station = access point (AP)
- Basic Service Set (BSS) (aka "cell") in infrastructure mode contains:
  - wireless hosts
  - access point (AP): base station
  - ad hoc mode: hosts only

Basic Service Set (BSS)

- The BSS without an AP is a stand-alone network and cannot send data to other BSSs.
  - called an ad hoc BSS
- A BSS with an AP is called an infrastructure BSS

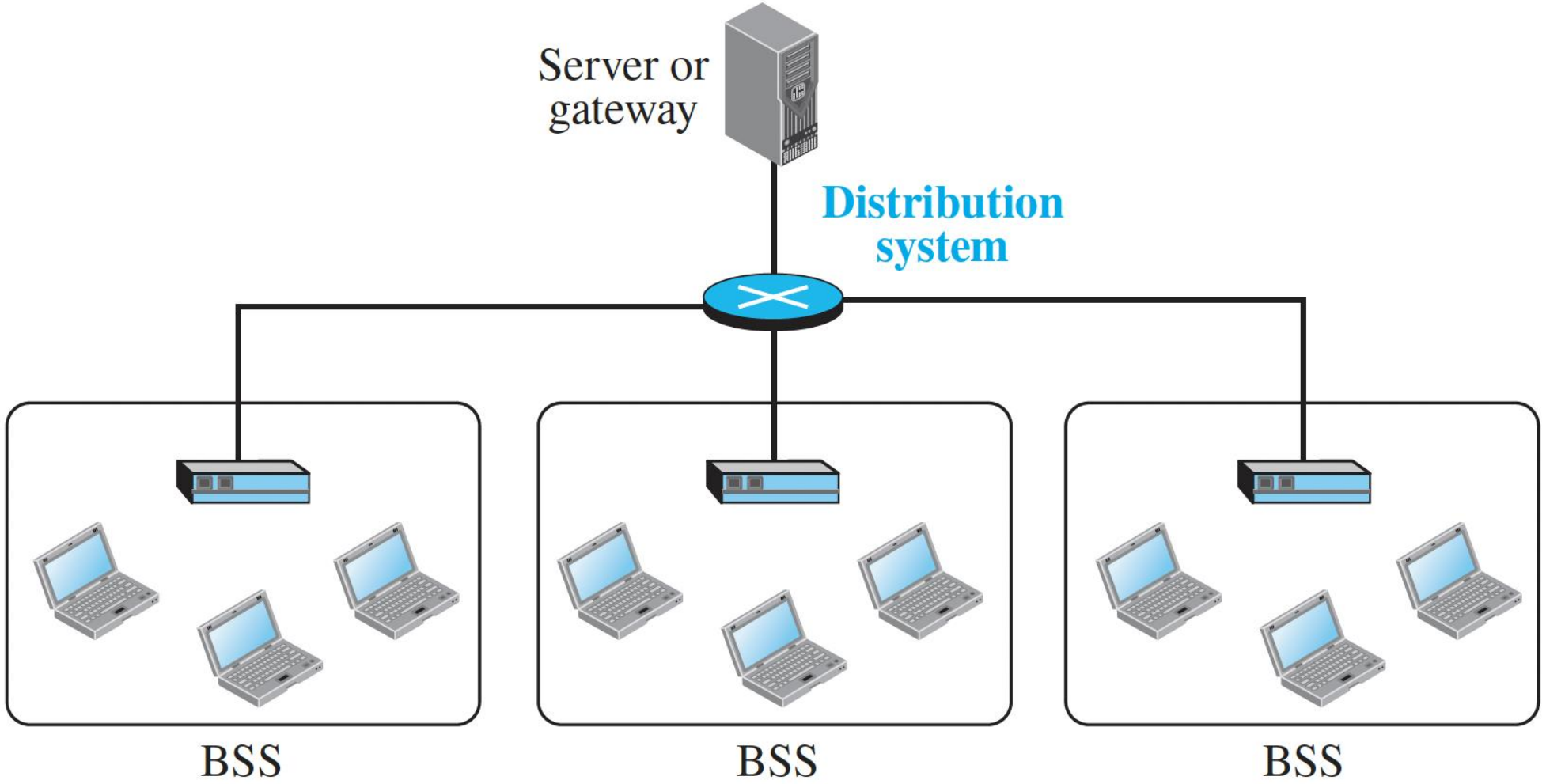


Ad hoc BSS

Infrastructure BSS

Extended Service Set (ESS)

- An ESS is made up of two or more BSSs with APs



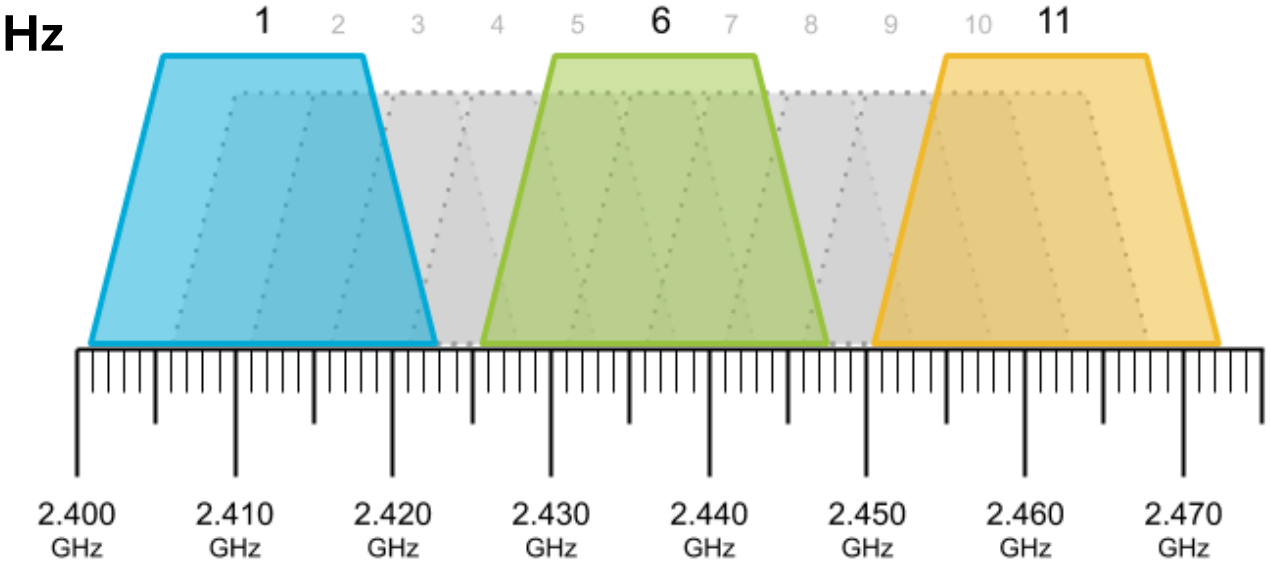
Extended Service Set (ESS)

- The BSSs in an ESS are connected through a distribution system, which is a wired or a wireless network
- Uses two types of stations: mobile and stationary
  - Mobile stations are normal stations inside a BSS
  - Stationary stations are AP stations that are part of a wired LAN
- When BSSs are connected, the stations within reach of one another can communicate without the use of an AP
- However, communication between a station in one BSS and an outside BSS occurs via the AP

802.11: Channels

- spectrum divided into channels at different frequencies
  - Channels overlap
- AP admin chooses frequency for AP
- interference possible: channel can be same as that chosen by neighboring AP!

Example: 2.4 GHz

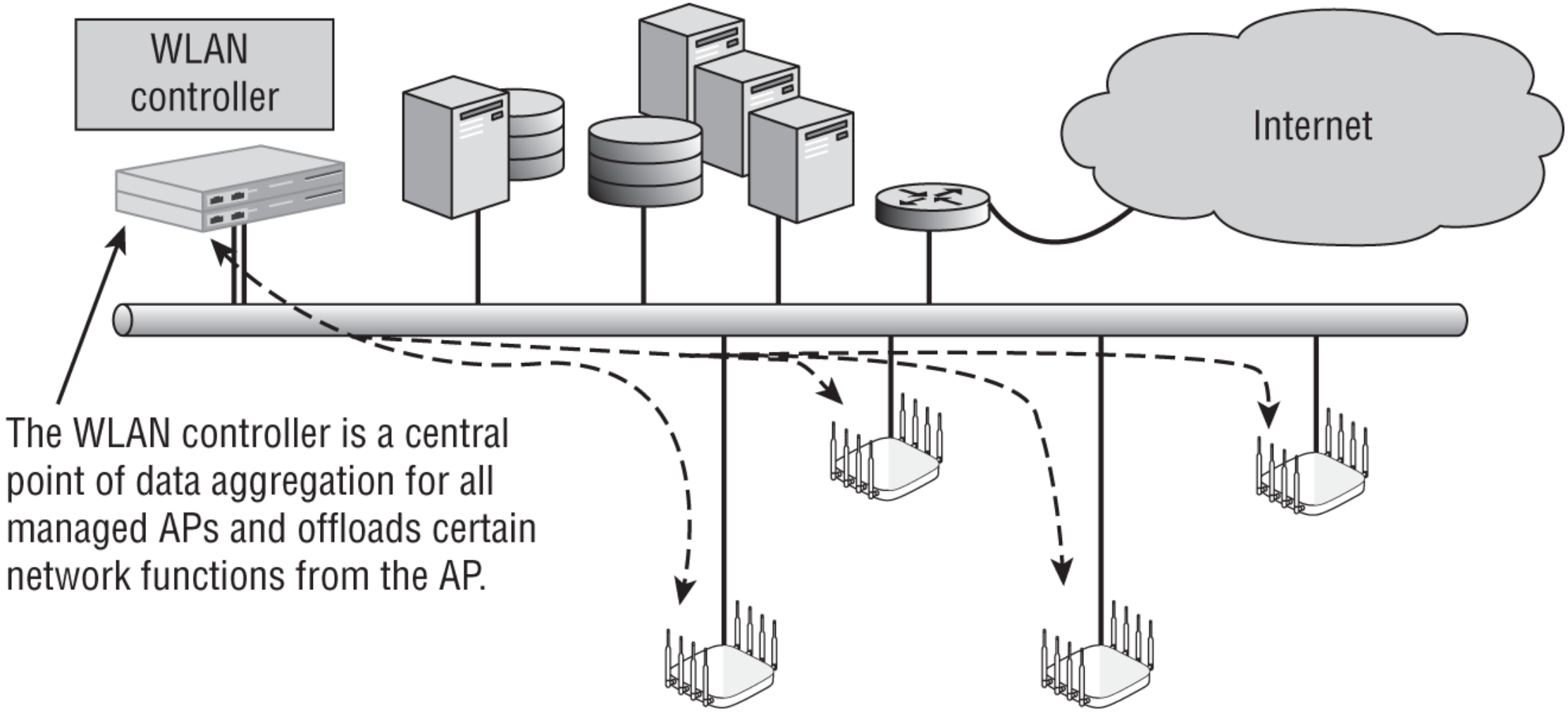


In the 2.4GHz band, only channels 1, 6 and 11 do not overlap

Centralized WLAN Architecture

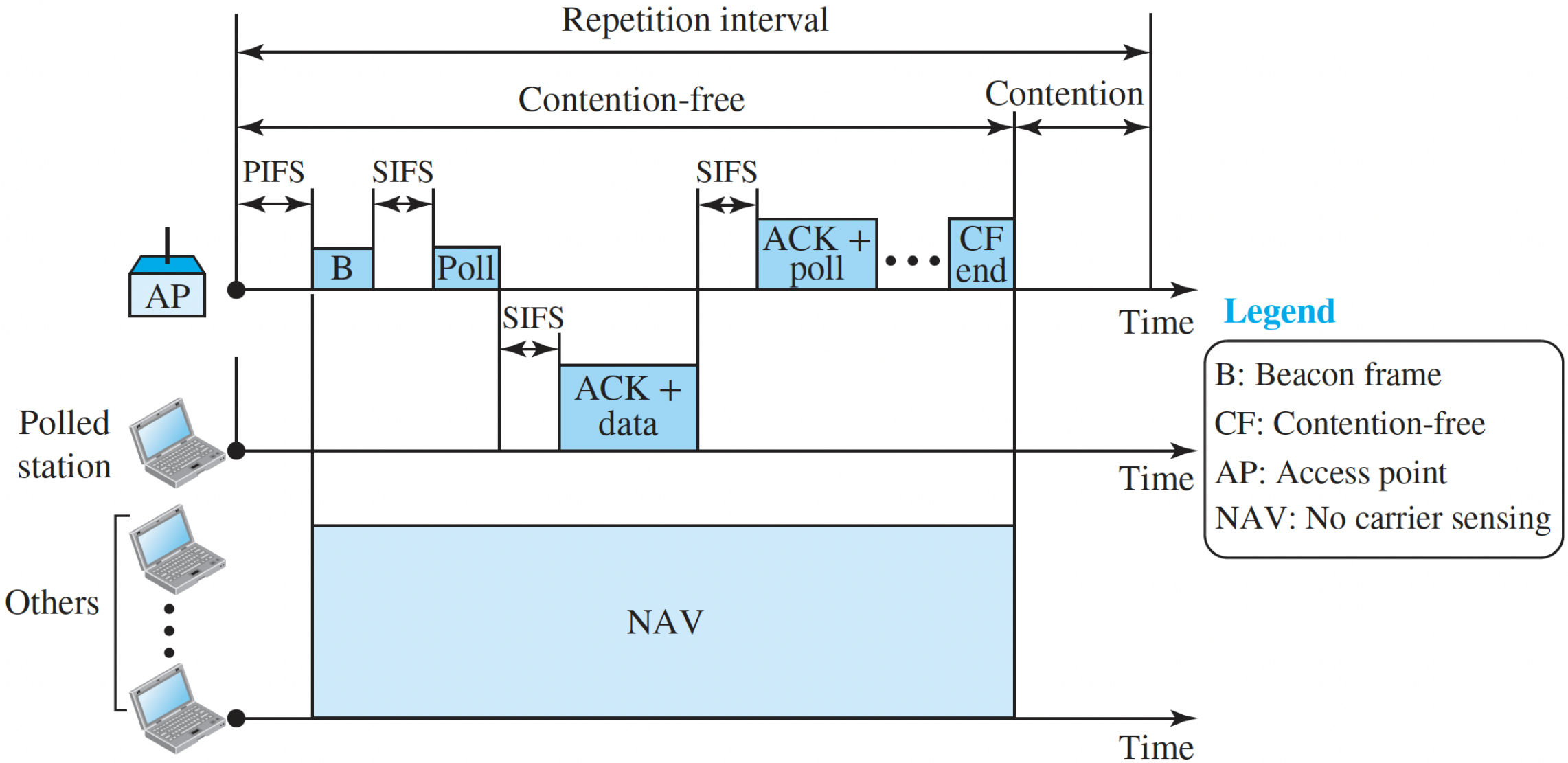
- In this approach, a central WLAN controller (WLC) that resides in the core of the network is used
- Autonomous APs are replaced with controller-based access points, also known as lightweight APs
- WLAN controller functions are three fold:
- Management: APs are configured and managed from the WLC
- Control: Adaptive RF, load balancing, roaming handoffs, and other mechanisms exist in the WLC
- Data: The WLC serves as a data distribution point for user traffic. Access points tunnel all user traffic to the WLC

Centralized WLAN Architecture



802.11 Medium Access Control (MAC)

- 802.11 is a wireless technology
- Transmission medium is shared by stations in a BSS
- If two or more transmit at the same time, a collision can occur
- Two protocols used for MAC
  - Distributed Co-ordinated Function (DCF) – all stations must contend for medium access (contention service)
  - Point Co-ordinated Function (PCF) – Access Point performs polling of stations, and grants them permission to transmit (contention-free service)
- DCF followed by PCF (called a superframe) is repeated


Repetition interval
Contention-free
Contention
PIFS
SIFS
SIFS
B
Poll
ACK + poll
CF end
AP
Time
SIFS
ACK + data
Polled station
Time
Legend
B: Beacon frame
CF: Contention-free
AP: Access point
NAV: No carrier sensing
Others
NAV
Time

IEEE 802.11: Inter Frame Space (IFS)

- Uses 4 different IFS values to provide a priority scheme
- SIFS (short IFS)
  - For all immediate response actions (ACK and CTS)
- PIFS (point coordination function IFS)
  - Used by the centralized controller in PCF scheme when issuing polls
- DIFS (distributed coordination function IFS)
  - Used as minimum delay to wait before sending a frame during DCF when stations contend for access
- EIFS (Extended IFS)
  - Used for error recovery
- SIFS < PIFS < DIFS < EIFS

IEEE 802.11: IFS values

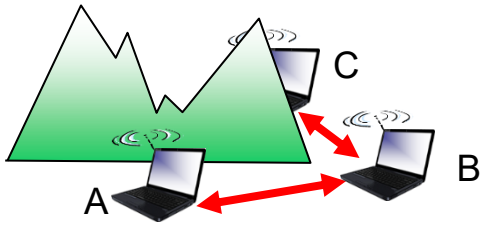
- Based on a parameter called slot time
  - Slot time = 9μs for 802.11n/ac/ax and 20μs for 802.11a/b/g

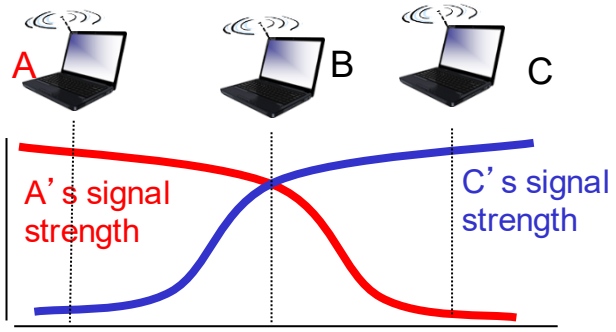
Parameter	802.11 a/b/g (2.4GHz)	802.11 n/ac/ax (5GHz)
Slot time	20	9
SIFS	10	16
PIFS = SIFS + slot time	30	25
DIFS = SIFS + 2 *slot time	50	34
EIFS	364	94

Some variations exist in 2.4GHz values

IEEE 802.11: multiple access

- Need to avoid collisions
- 802.11: CSMA - sense before transmitting
  - If a station is already transmitting, wait until it is over
- 802.11: *no* collision detection!
  - difficult to sense collisions: high transmitting signal, weak received signal due to fading
  - can't sense all collisions in any case: hidden terminal, fading
  - goal: *avoid collisions:* CSMA/CA (Collision Avoidance)





IEEE 802.11 MAC Protocol: CSMA/CA

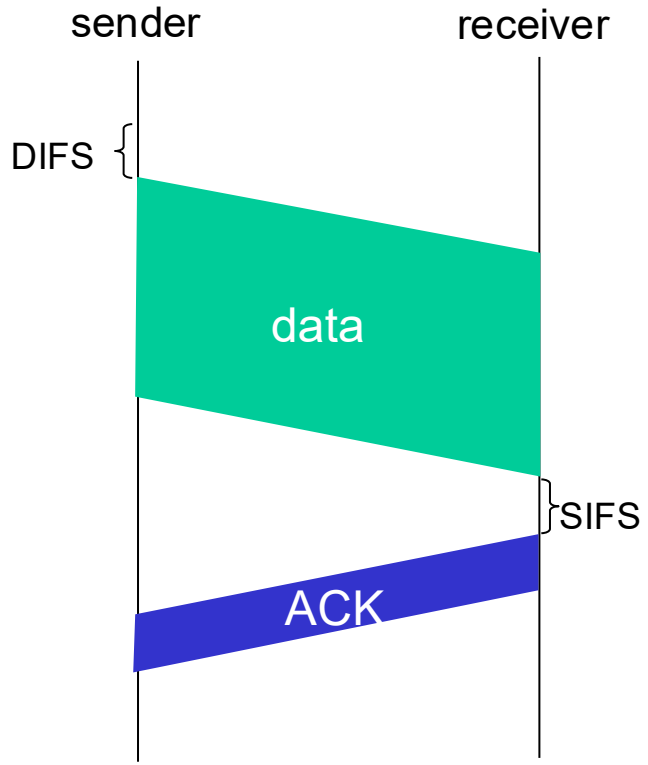
802.11 sender

1 if sense channel idle for **DIFS** then
transmit entire frame (no CD)

2 if sense channel busy then
start random backoff time
timer counts down while channel idle
transmit when timer expires
if no ACK, increase random backoff interval, repeat 2

802.11 receiver

if frame received OK
return ACK after **SIFS** (ACK needed to ensure that the frame was received successfully)



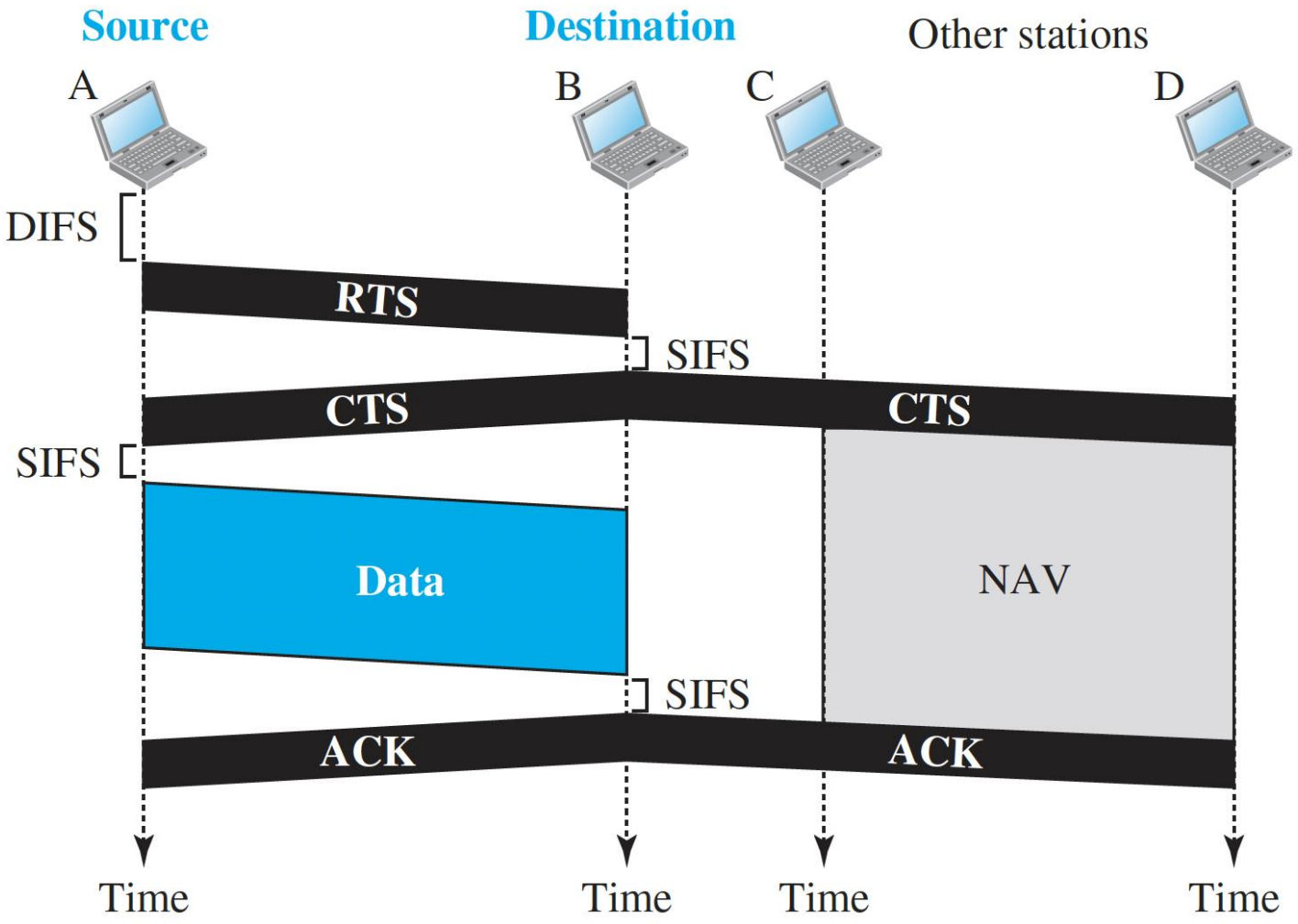
Basic access mode

Avoiding collisions (more)

idea: sender "reserves" channel use for data frames using small reservation packets

- sender first transmits *small* request-to-send (RTS) packet to receiver using CSMA
  - RTSs may still collide with each other (but they're short)
- receiver broadcasts clear-to-send CTS in response to RTS
- CTS heard by all nodes
  - sender transmits data frame
  - other stations defer transmissions

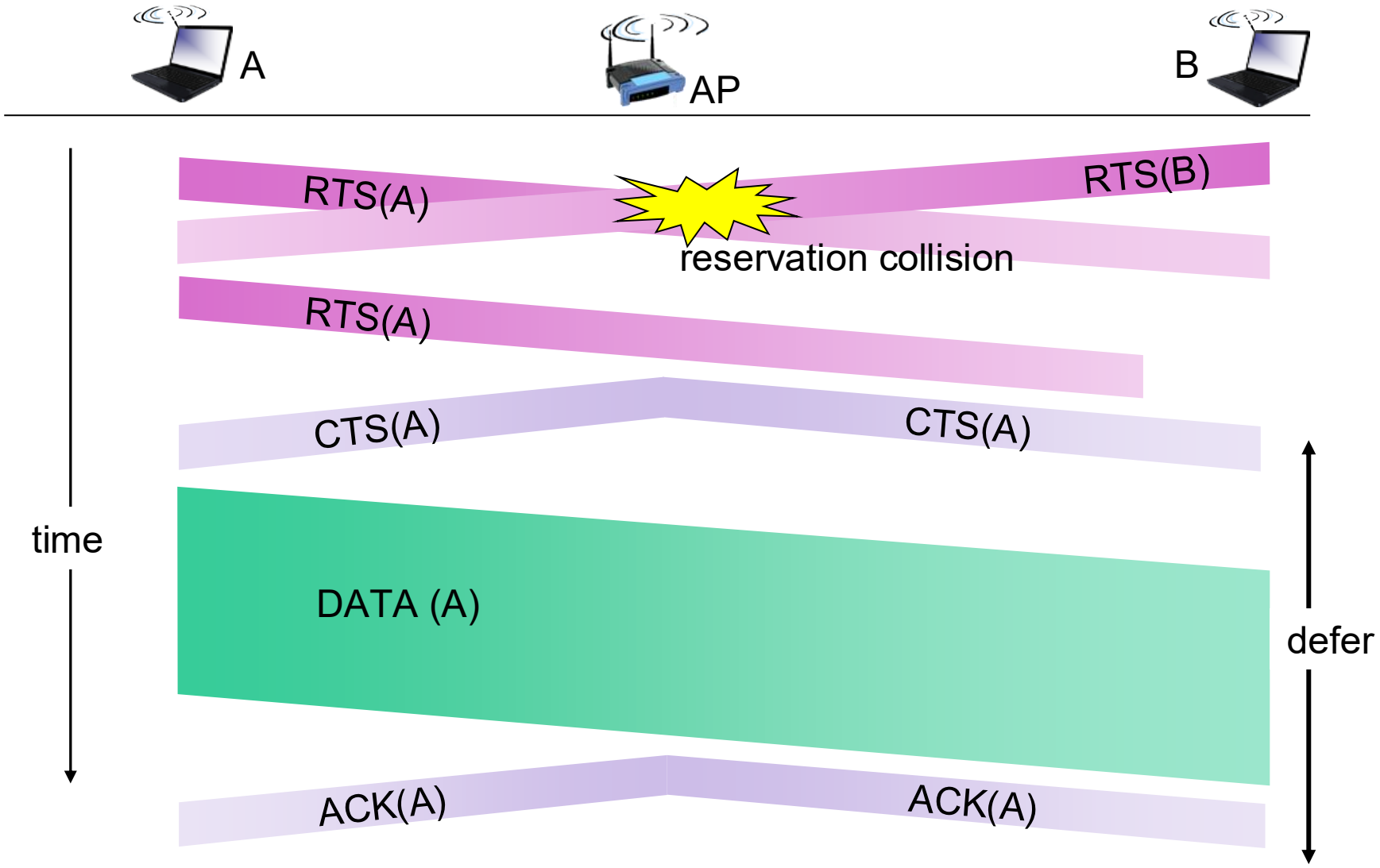
Four Frame Exchange



1. Source issues a Request to Send (RTS) frame
2. Destination responds with Clear to Send (CTS)
3. After receiving CTS, source transmits data
4. Destination responds with ACK

RTS/CTS access mode

Collision Avoidance: RTS-CTS exchange



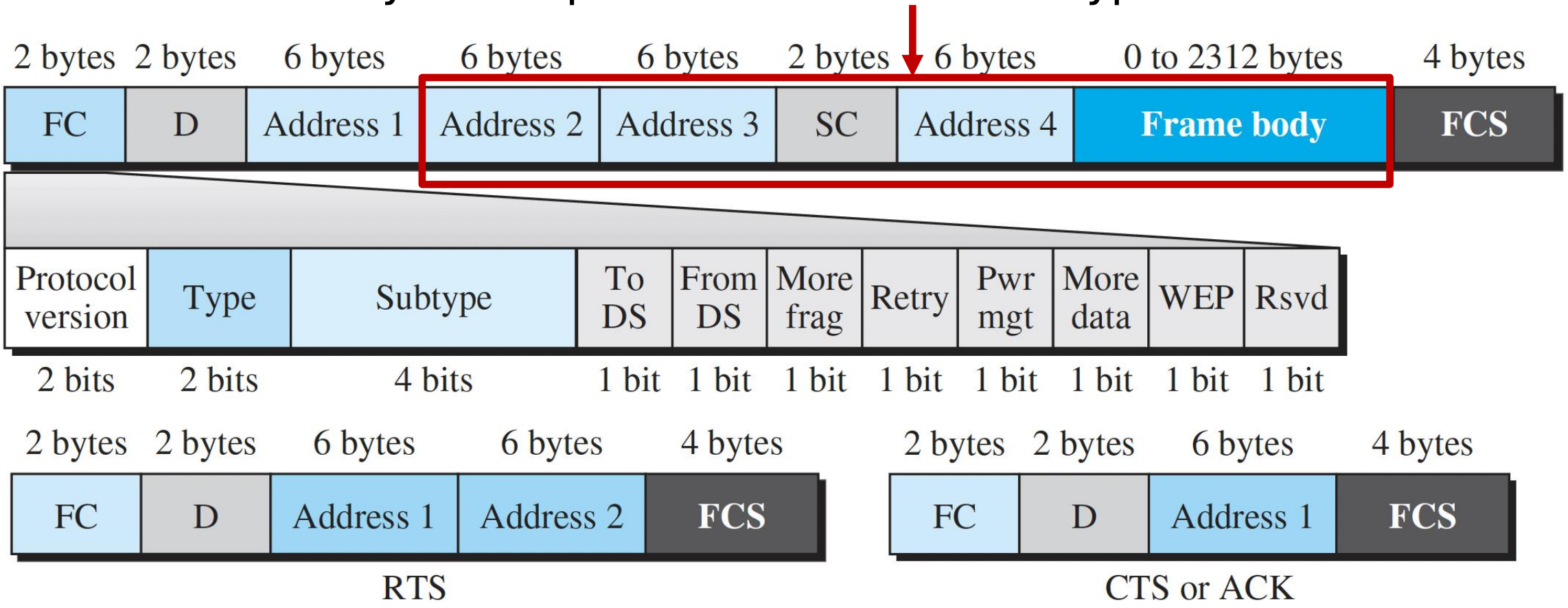
A collision occurs when both A and B send RTS frames at the same time.

A tries again and succeeds

Sender is A
Receiver is AP

IEEE 802.11 frame format

- An 802.11 frame consists of nine fields
- Some fields may not be present in some frame types

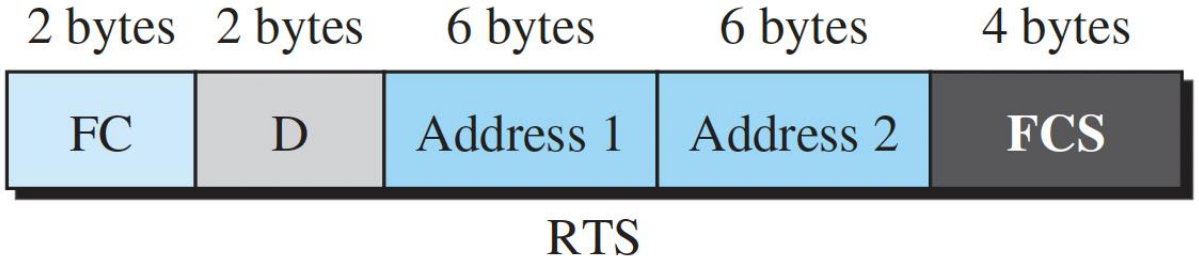


IEEE 802.11 frame format

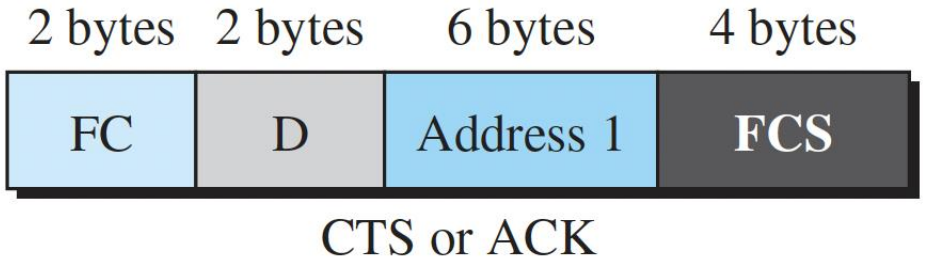
- Frame control (FC) – 2 bytes
  Defines the type of frame and some control information
- D – 2 bytes
  Defines the duration of the transmission that is used to set the value of NAV.
  In one control frame, it defines the ID of the frame.
- Addresses – 6 bytes each
  There can be up-to four address fields
  The meaning of each address field depends on the value of the To DS and From DS subfields.
- Sequence control (SC) – 2 bytes
  The first 4 bits define the fragment number; the last 12 bits define the sequence number, which is the same in all fragments.
- Frame body – 0 to 2312 bytes
  Contains information based on the type and the subtype defined in the FC field
- FCS – 4 bytes
  Contains a CRC-32 error-detection sequence.

IEEE 802.11 frame types

- Management Frames (Type = 00)
  Used for managing communication between stations and access points
  Request, response, reassociation, dissociation, and authentication etc.
- Control Frames (Type = 01)
  Used for accessing the channel and acknowledging frames
- Data frames (Type = 10)
  Used for carrying data and control information
- Extension frames (Type = 11)
  A flexible frame format, currently used only with 802.11ad and 802.11ah)



RTS



CTS or ACK

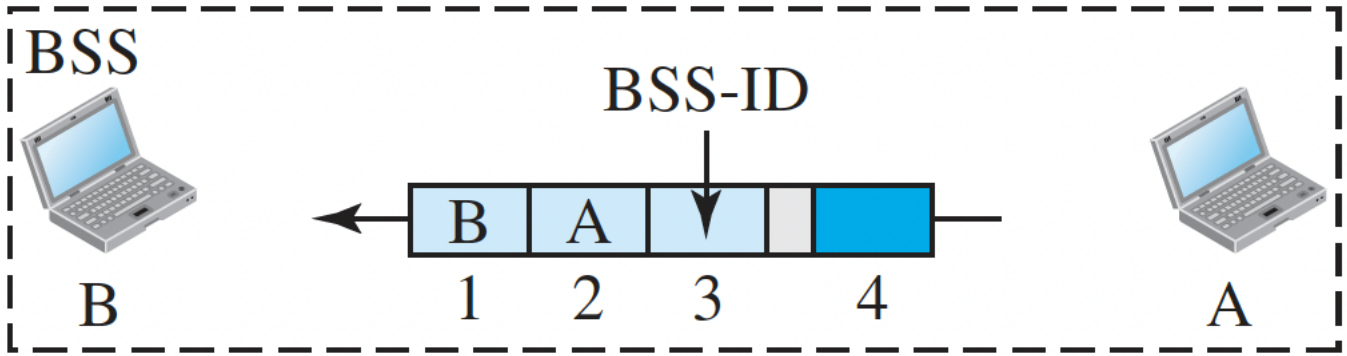
Addressing mechanism

To DS	From DS	Address 1	Address 2	Address 3	Address 4
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

- BSS ID is the same as the MAC address of the AP in the BSS
- Address 1 is always the address of the next device that the frame will visit
- Address 2 is always the address of the previous device that the frame has left
- Address 3 : Address of the final destination station if it is not defined by address 1 or the original source station if it is not defined by address 2
- Address 4 is the original source when the distribution system is also wireless

Addressing: case 1

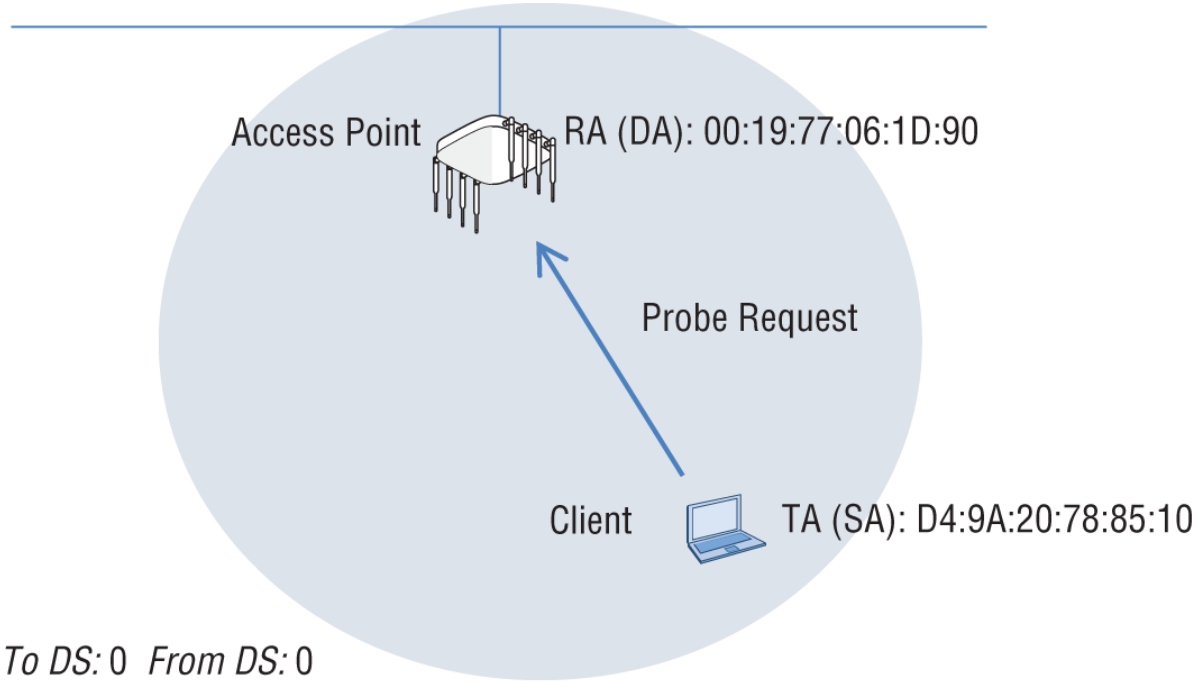
- In this case, To DS = 0 and From DS = 0
- This means that the frame is not going to a distribution system (To DS = 0) and is not coming from a distribution system (From DS = 0)
- The frame is going from one station in a BSS to another without passing through the distribution system.



a. Case 1

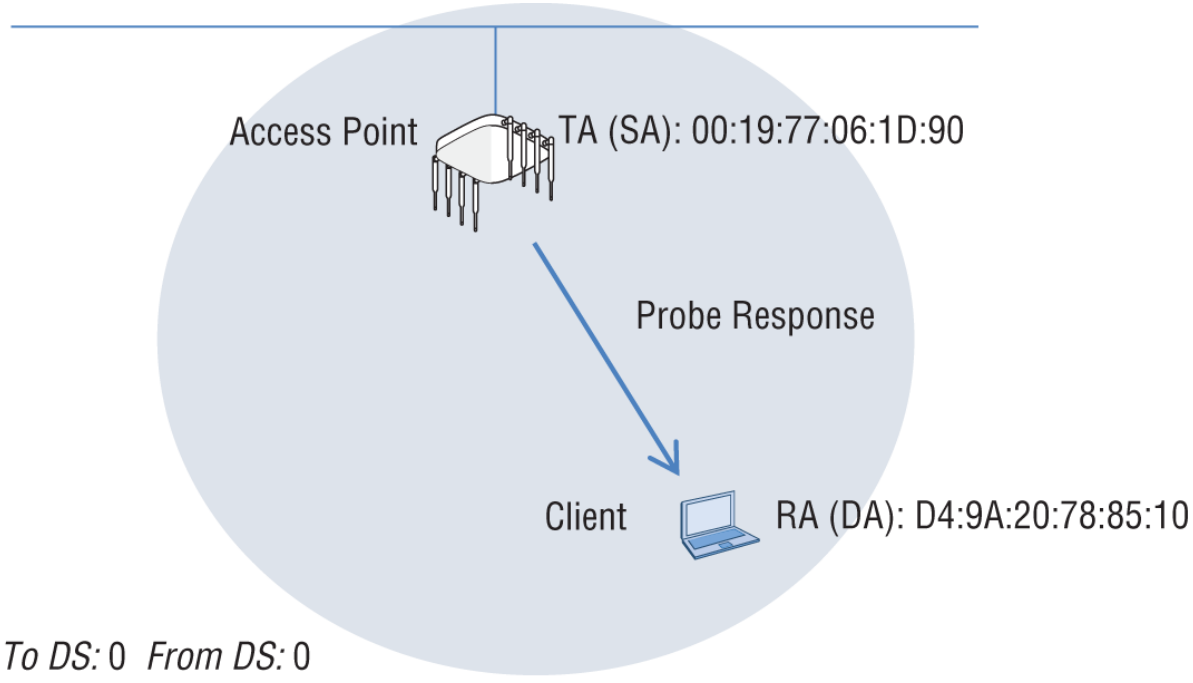
Addressing: case 1 (ctd)

- All Management and Control frames also use To DS = 0 and From DS = 0



To DS: 0 *From DS:* 0

Address #1: RA (DA): 00:19:77:06:1D:90
Address #2: TA (SA): D4:9A:20:78:85:10
Address #3: BSSID: 00:19:77:06:1D:90

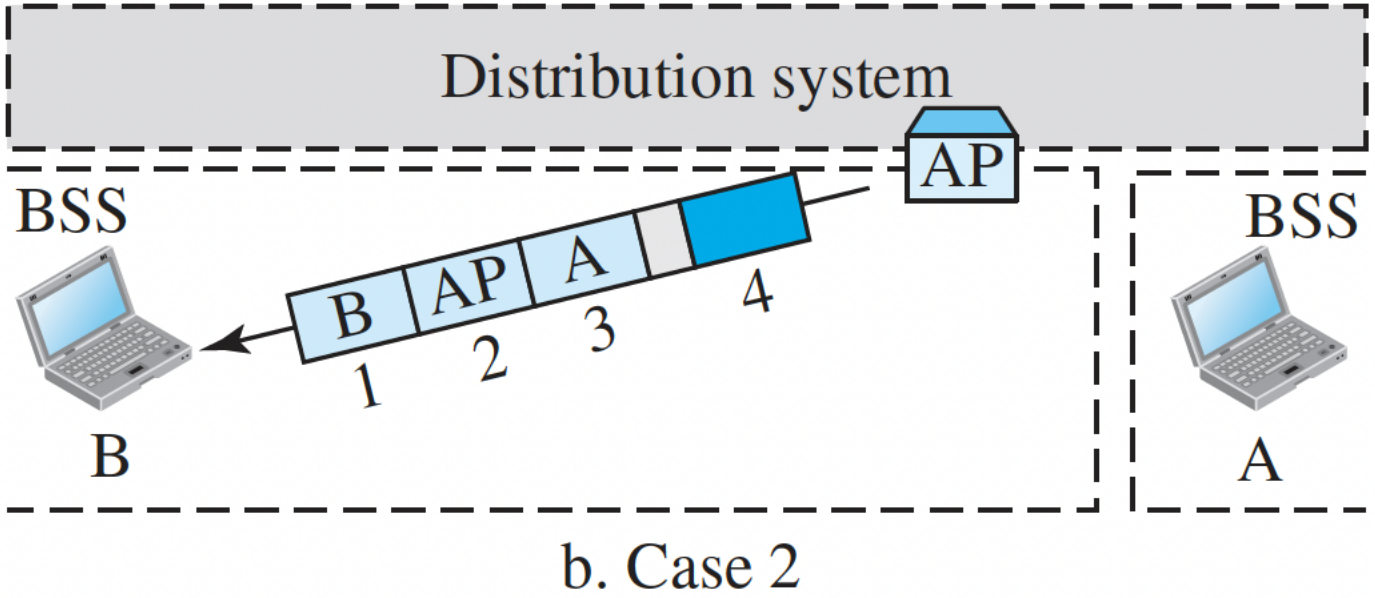


To DS: 0 *From DS:* 0

Address #1: RA (DA): D4:9A:20:78:85:10
Address #2: TA (SA): 00:19:77:06:1D:90
Address #3: BSSID: 00:19:77:06:1D:90

Addressing: case 2

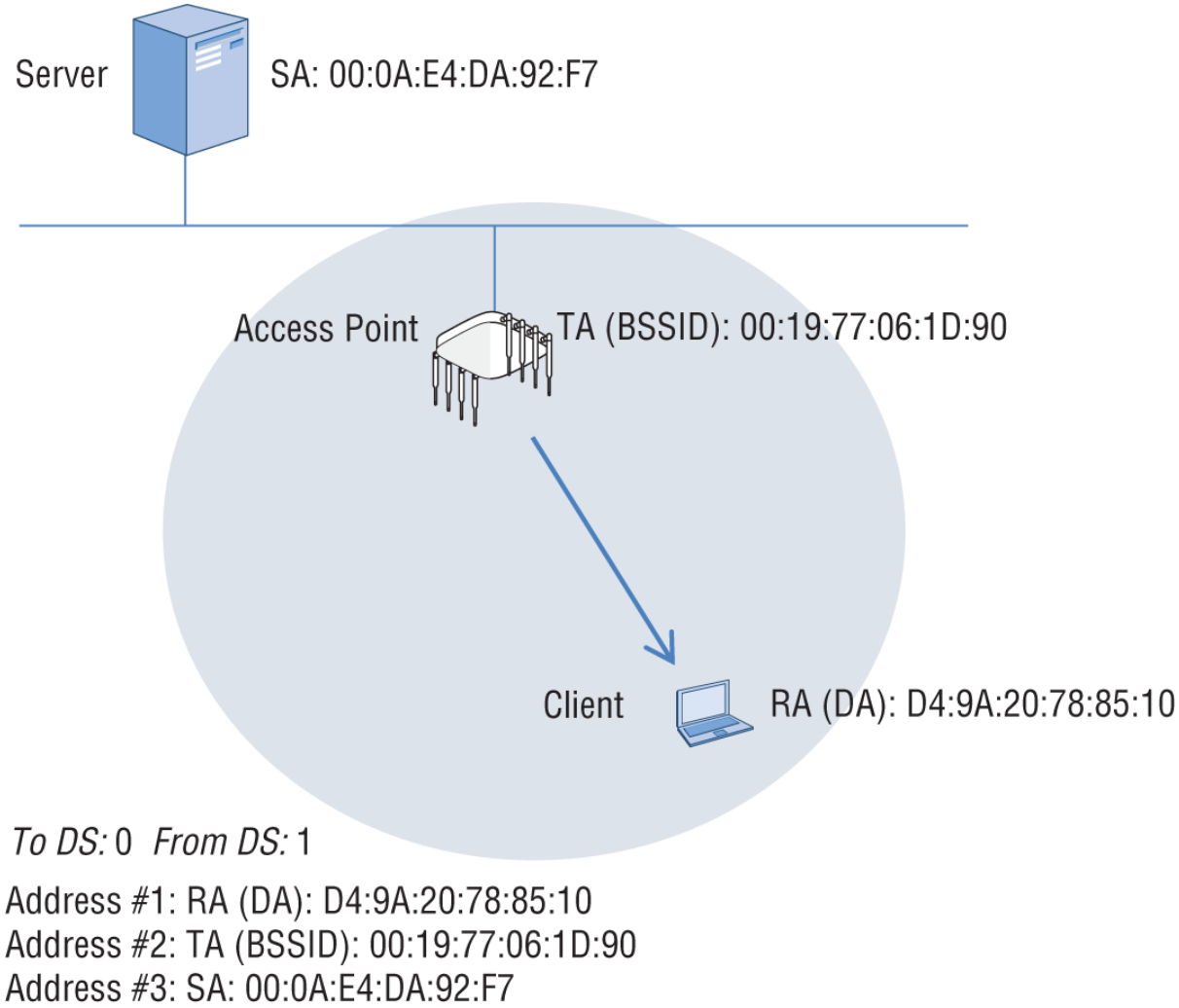
- In this case, To DS = 0 and From DS = 1
- This means that the frame is coming from a distribution system (From DS = 1)
- The frame is coming from an AP and going to a station
- Note that address 3 contains the original sender of the frame (in another BSS)



b. Case 2

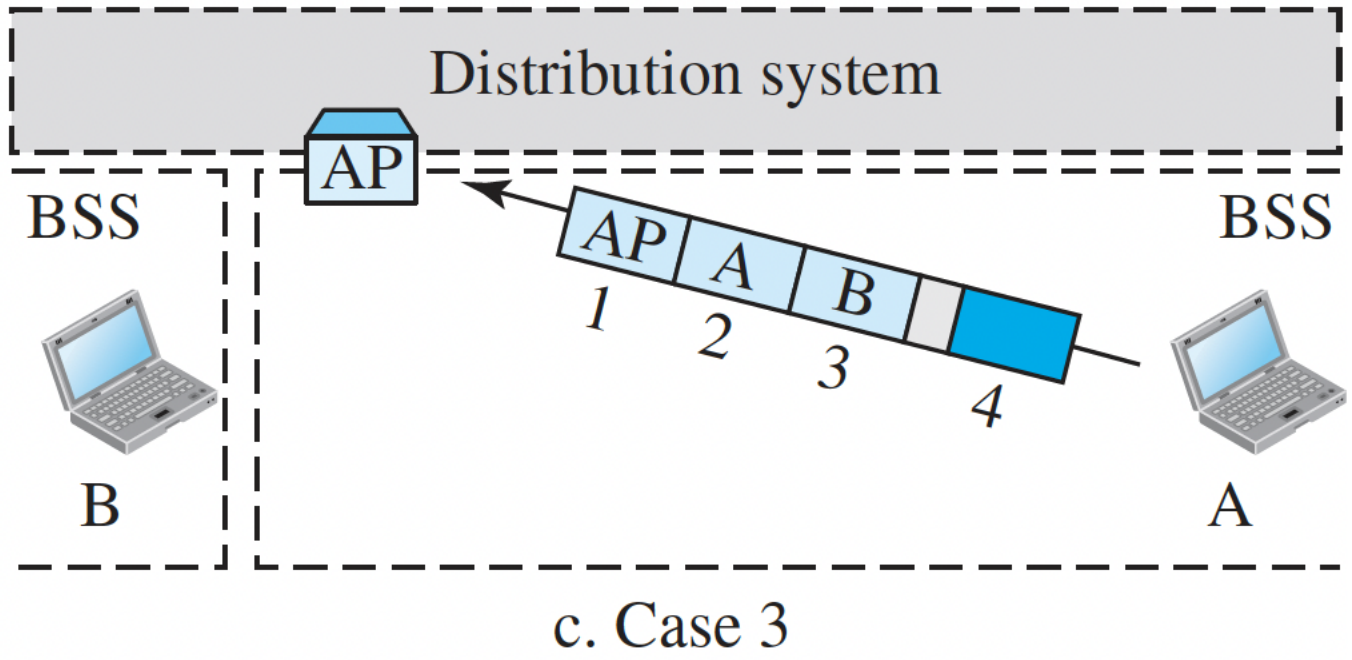
Addressing: case 2 (ctd)

- Frames from a device in the wired network to a wireless station will also have To DS = 0 and From DS = 1



Addressing: case 3

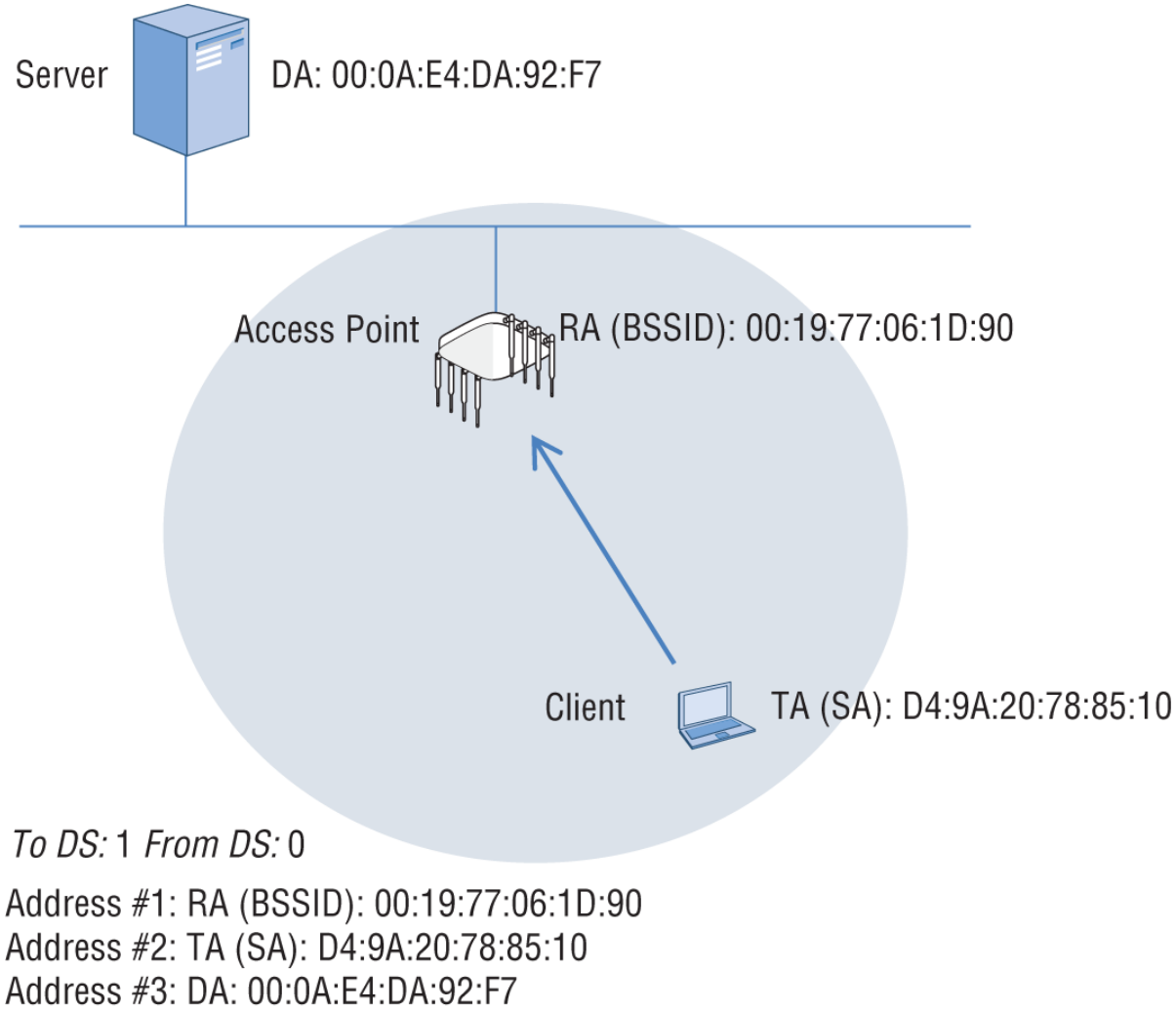
- In this case, To DS = 1 and From DS = 0
- This means that the frame is going to a distribution system (To DS = 1)
- The frame is going from a station to an AP
- ACK is sent to the original station
- Address 3 contains the final destination of the frame in the distribution system.



c. Case 3

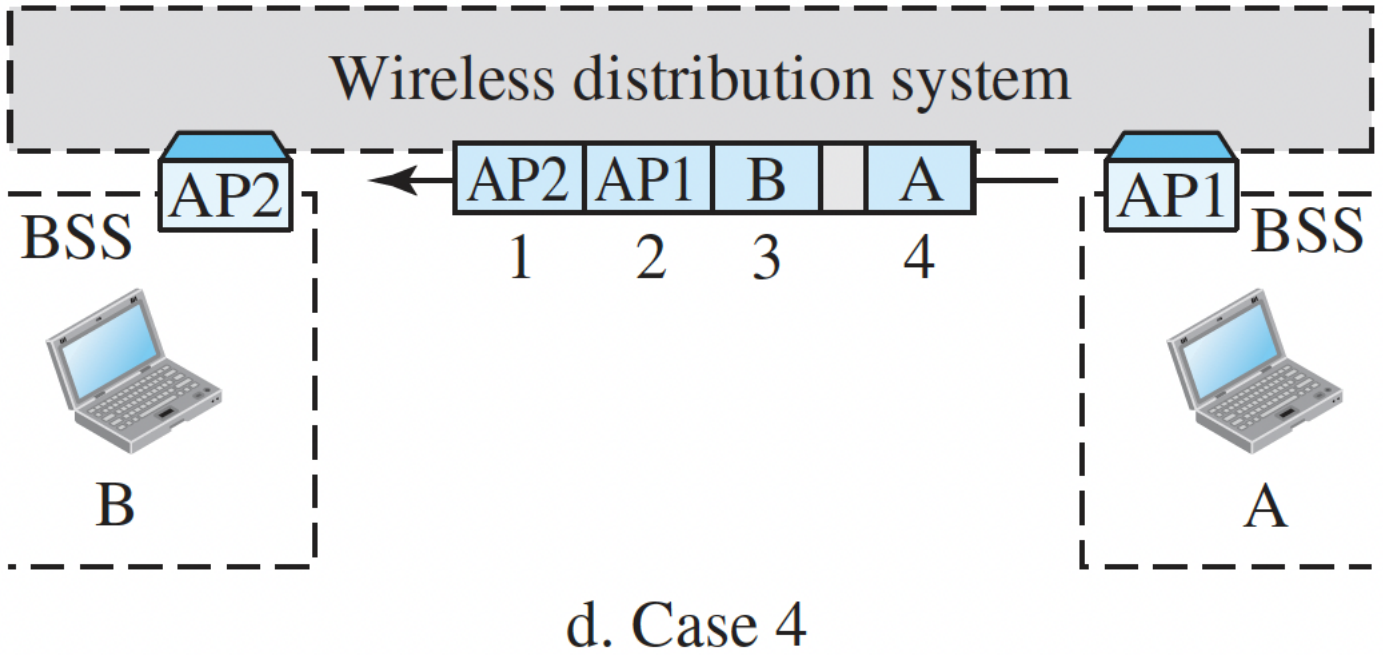
Addressing: case 3 (ctd)

- Frames from a wireless station to a host in the wired network will also have To DS = 1 and From DS = 0



Addressing: case 4

- In this case, To DS = 1 and From DS = 1
- This is the case in which the distribution system is also wireless
- The frame is going from one AP to another AP in a wireless distribution system
- Here, we need four addresses
  - Original sender
  - Final destination
  - Two intermediate APs

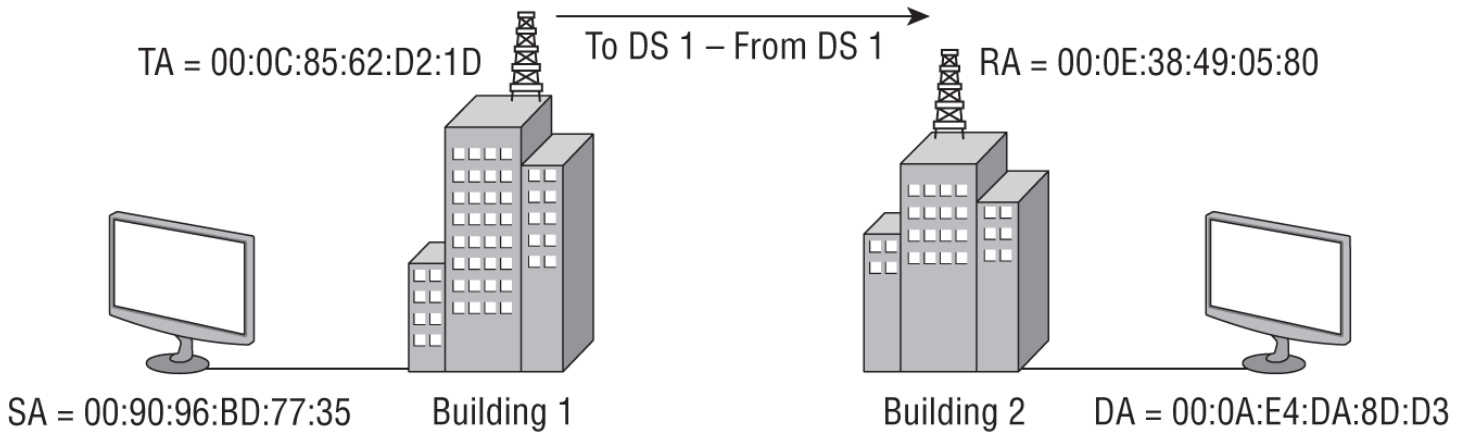


d. Case 4

Addressing: case 4 (ctd)

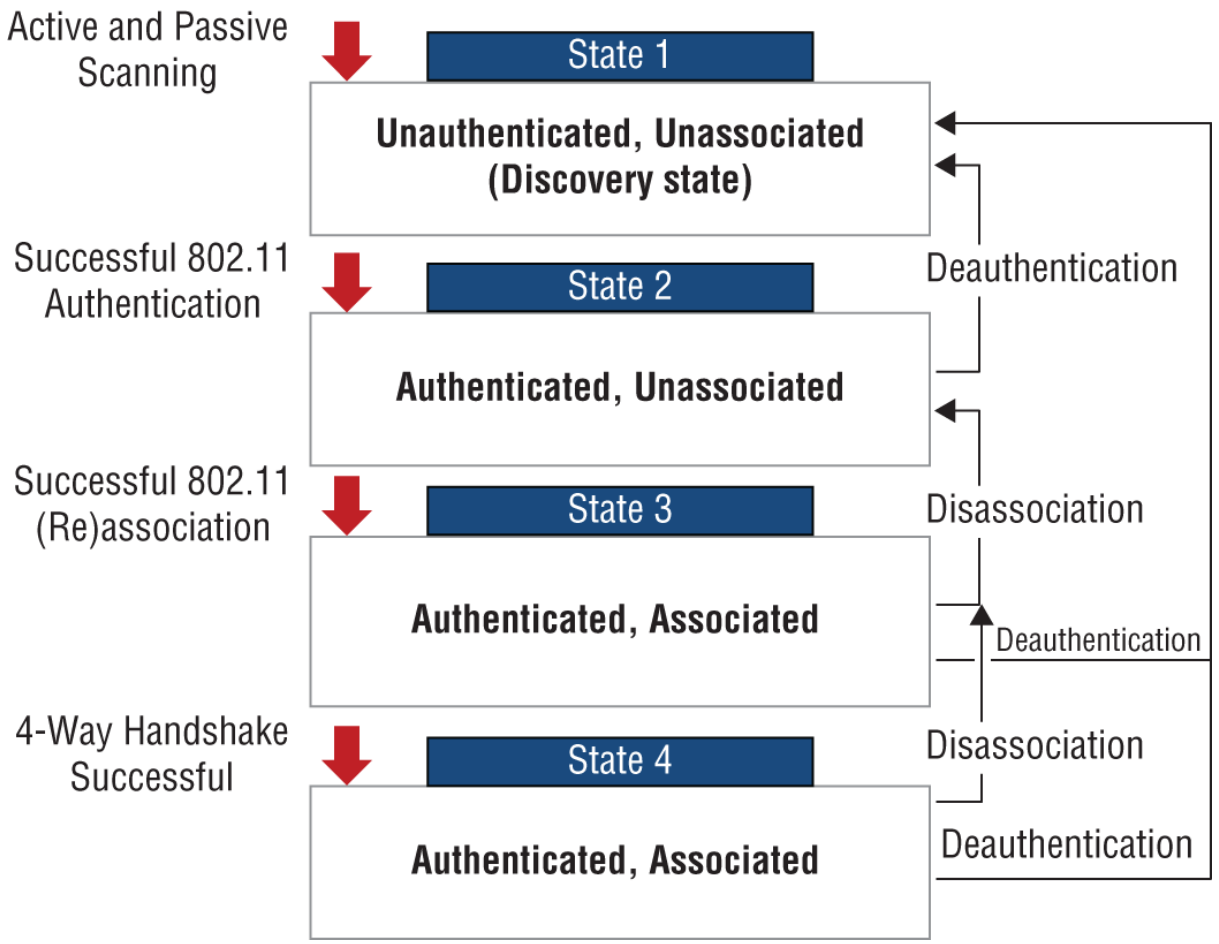
- Wireless bridges used in a wired network will also use To DS = 1 and From DS = 1

802.11 MAC Header	
Version:	*0*
Type:	*%10 Data*
Subtype:	*%1000 QoS Data*
Frame Control Flags:	*%00000011*
Duration:	*44 Microseconds*
Receiver:	*00:0E:38:49:05:80*
Transmitter:	*00:0C:85:62:D2:1D*
Destination:	*00:0A:E4:DA:8D:D3*
Source:	*00:90:96:BD:77:35*
Seq Number:	*982*
Frag Number:	*0*



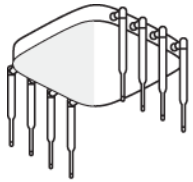
802.11 client state machine

- The 802.11 standard defines four states of client connectivity
- 802.11 management frames are exchanged between a client station and an AP, as the client transitions between the four states to establish layer 2 connectivity
- State 1: Initial start state, unauthenticated and unassociated
- State 2: Authenticated, not associated
- State 3: Authenticated and associated (pending RSN authentication)
- State 4: Authenticated and associated

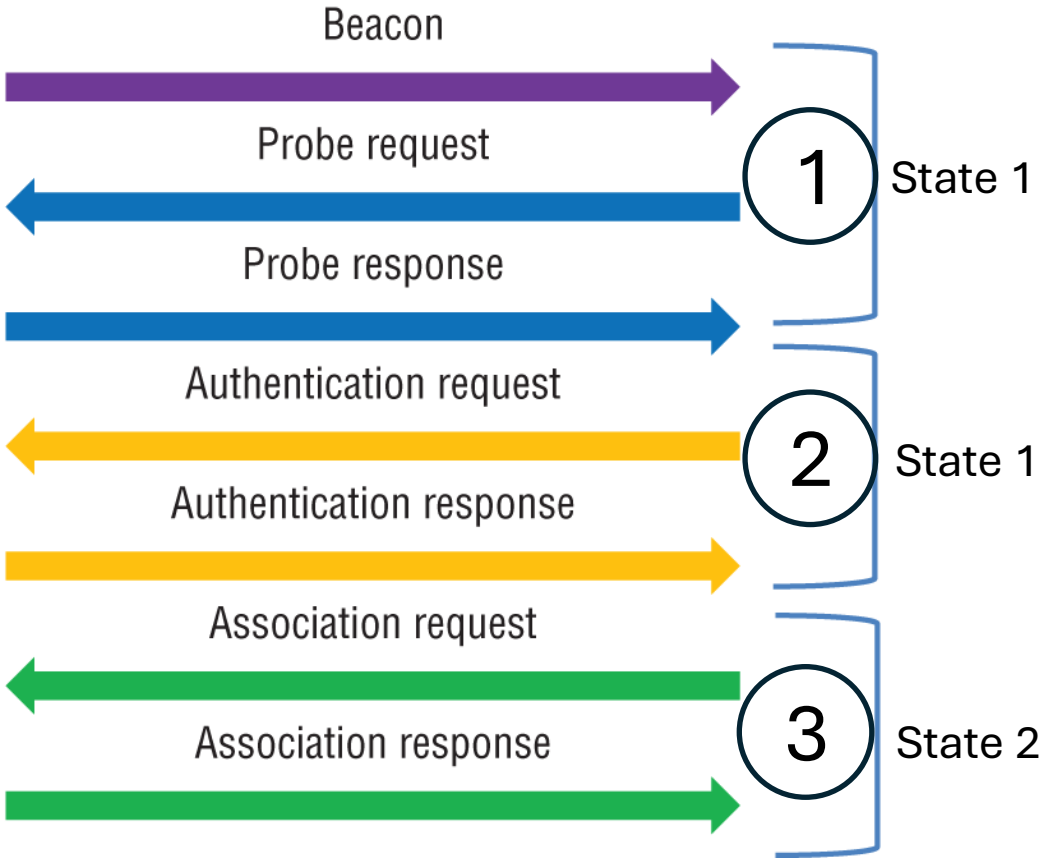


Authentication & Association

1. Active / Passive scanning
2. Open System Authentication (a.k.a. Null Authentication) with the AP
3. Association with the AP







802.11: Association

- arriving host: must associate with an AP
- scans channels, listening for *beacon frames*
- Beacon frames contain
  - name of the wireless network (SSID)
  - MAC address of the AP (BSSID)
- Host selects AP to associate with and joins the network
- Then may perform authentication
- Then get IP address in AP's subnet



Beacon

- Beacons are essentially the heartbeat of the wireless network
- Broadcast by the AP of a basic service set
- Client stations transmit beacons only when participating in an independent basic service set (IBSS), also known as ad hoc mode
- Each beacon contains a time stamp, which client stations use to keep their clocks synchronized with the AP
- Because so much of successful wireless communication is based on timing, it is imperative that all stations be in sync with each other
- Over 75 fixed fields and information elements can be in a Beacon frame depending on the operational modes of the access point
- Beacons are transmitted at a targeted time of every 102.4 milliseconds
  - About 10 beacons per second

Beacon frame fields

- **Time Stamp**: Synchronization information
- **Beacon Interval**: Number of time units (TUs) between target beacon transmission times (TBTTs)
- **SSID**: Logical WLAN name
- **Data Rates**: Basic and supported rates
- **Service Set capabilities**: Extra BSS or IBSS parameters
- **Country code**: Country code identifier for spectrum regulatory purposes
- **Channel information**: Channel used by the AP or IBSS
- … and a lot of other fields

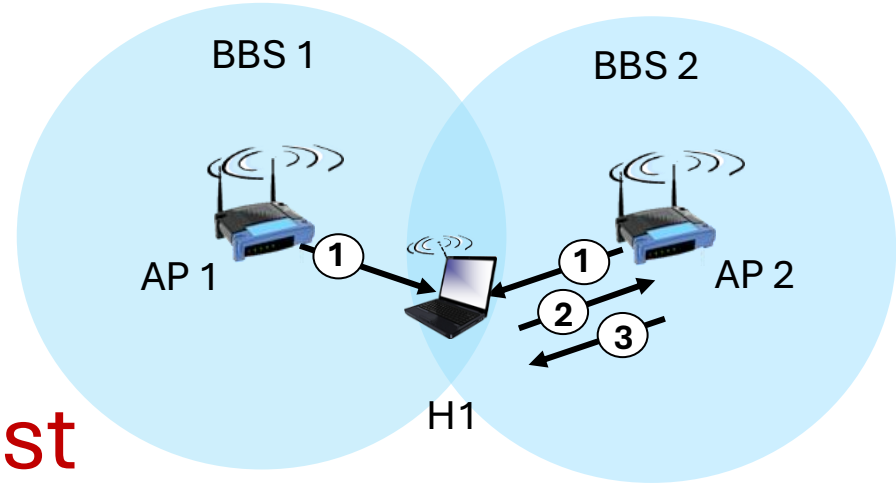
Basic and supported rates

- Specific data rates can be configured for any AP as required rates or basic rates
- An AP will transmit all management frames at the lowest configured basic rate
- Data frames can be transmitted at much higher supported data rates
- For a client station to successfully associate with an AP, the station must be capable of communicating by using the configured basic rates that the AP requires
- If the client station is not capable of communicating with **all** of the basic rates, the client station will not be able to associate with the AP and will not be allowed to join the BSS
- In addition to the basic rates, the AP defines a set of supported rates, and advertised by the AP in the beacon frame and probe responses
- A client station **does not** have to support all of the supported rates of the AP

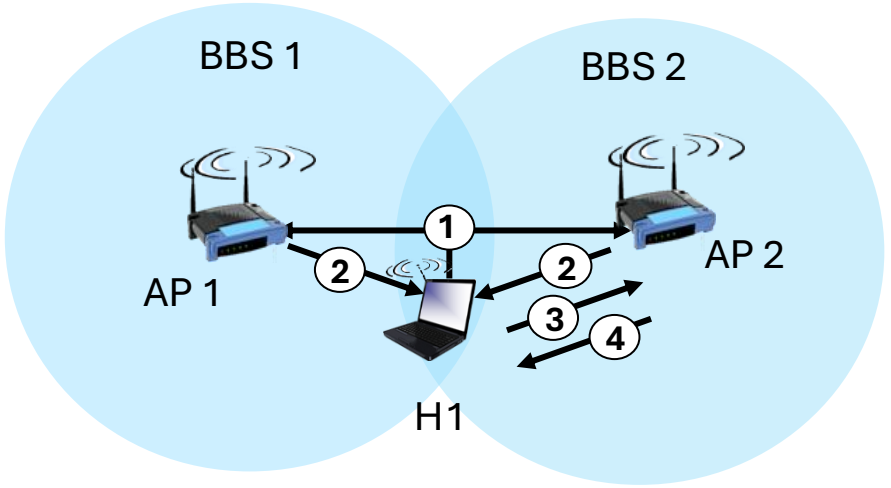
802.11: passive scanning

- In passive scanning hosts just listen to the Beacon frames sent by access points
- Then join the network by sending an Association Request frame to the selected AP
- AP responds via an Association Response frame
- Example:

  (1)Beacon frames sent from APs
  (2)Association Request frame sent: H1 to selected AP
  (3)Association Response frame sent from selected AP to H1



802.11: active scanning



- In active scanning, hosts send a Probe Request
  - Null Probe Request: does not specify an SSID
  - Directed Probe Request: specifies an SSID
- Matching APs will respond with a Probe Response
  - Null Probe Request: All APs receiving the request will respond
  - Directed Probe Request: Only the APs in the specified SSID will respond
- Probe Response is similar to a Beacon
- The host will select an AP and join the network
- Example:

  (1)Probe Request frame broadcast from H1
  (2)Probe Response frames sent from APs
  (3)Association Request frame sent: H1 to selected AP
  (4)Association Response frame sent from selected AP to H1

Authentication

- Authentication is the first of two steps required to connect to an 802.11 BSS
- Uses Authentication management frame
- The original 802.11 standard defined two different methods of authentication
- Shared Key authentication uses Wired Equivalent Privacy (WEP) to authenticate client stations and requires that a static WEP key be configured on both the station and the access point
- Because WEP is an outdated security method, Shared Key authentication is not used anymore
- Open System authentication provides authentication without performing any type of client verification (considered a null authentication)
- It is essentially an exchange of hellos between the client and the AP
- After successful authentication, the client moves to state 2 of state machine

Association

- After the station has authenticated with the AP, the next step is for it to associate with the AP
- Successful association means that the client station has established layer 2 connectivity with the AP and joined the BSS
- The client station sends an Association Request management frame to the AP, seeking permission to join the BSS
- The AP sends an Association Response management frame to the client, either granting or denying permission to join the BSS
- An association response frame includes an association identifier (AID), a unique association number given to every associated client
- After successful association, the client station moves to state 3 of the 802.11 state machine

Review questions

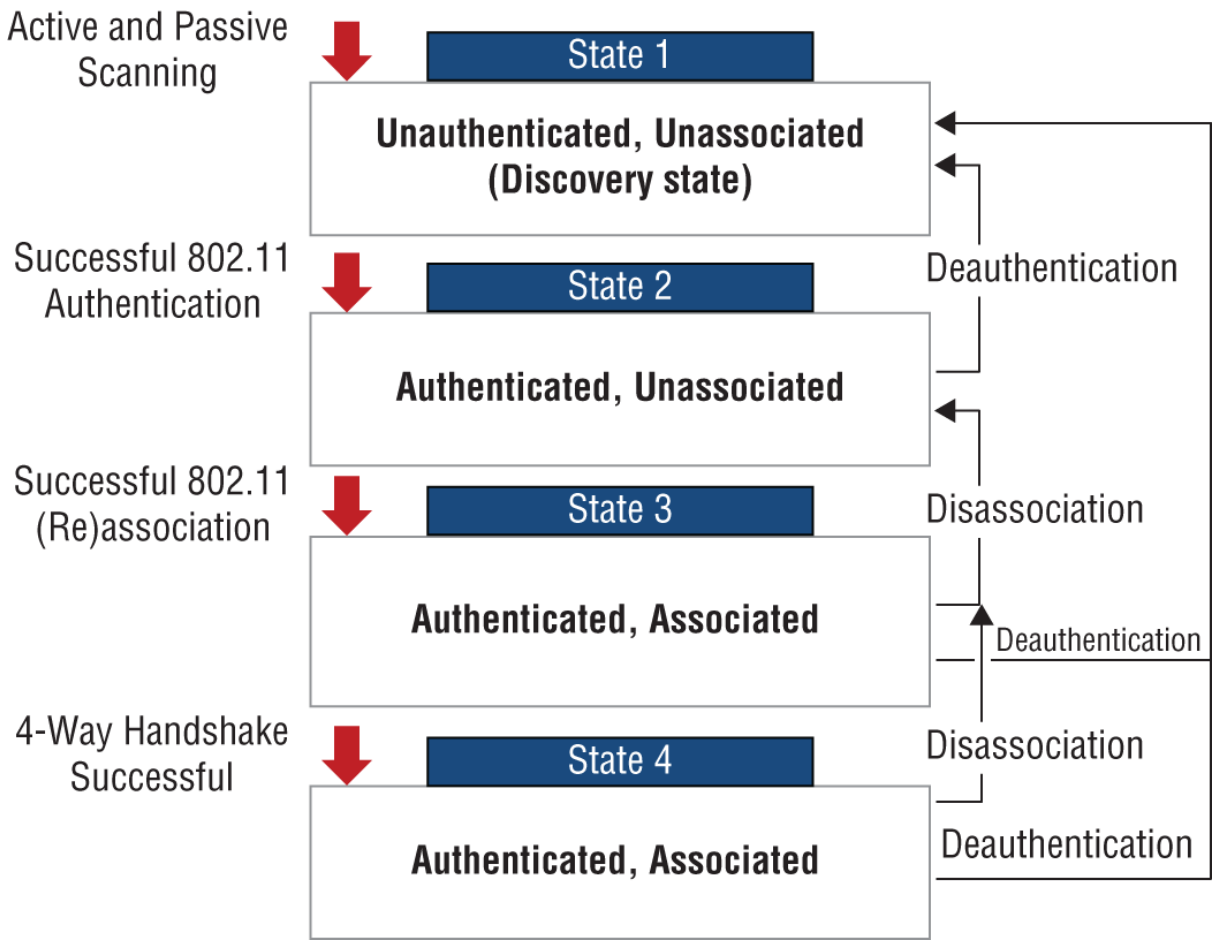
- What is a BSS?
- What is the difference between an ad-hoc BSS and an infrastructure BSS?
- What is an ESS?
- What is a distribution system (DS)?
- What is the difference between CSMA/CD and CSMA/CA?
- What is the difference between active and passive scanning?
- What are the four phases in a client joining an 802.11 wireless network?

References

- Chapter 6 – The Link Layer and LANs
  James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach, (8th Edition), Pearson, 2020
- Chapter 13 – Wireless LANs
  William Stallings, Data and Computer Communications (10th Edition), Pearson, 2014

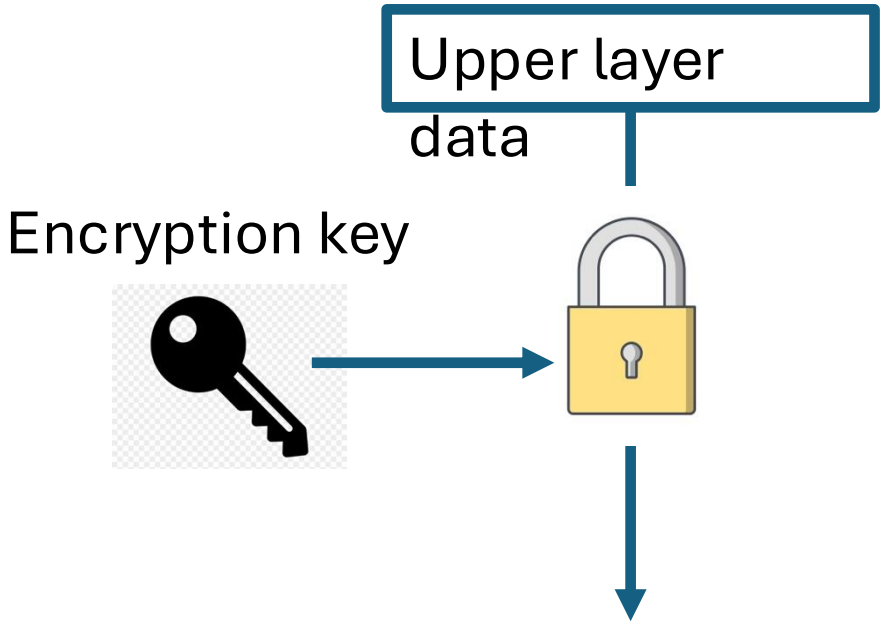
RSN security

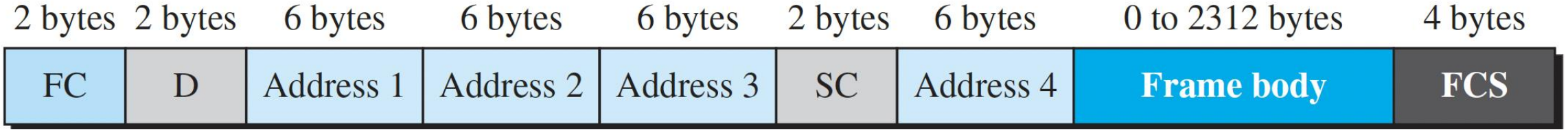
- If Robust Security Network (RSN) authentication is NOT used, a client station in state 3 can move beyond layer 2, request an IP address, and begin upper-layer communications
- If RSN authentication is used, the client station has still not joined the BSS
- The client is associated, but RSN authentication is pending
- RSN authentication uses PSK, SAE or 802.1X/EAP authentication
- PSK – Pre Shared Key (PSK)
- SAE – Simultaneous Authentication of Equals
- EAS – Extensible Authentication Protocol



WLAN Encryption

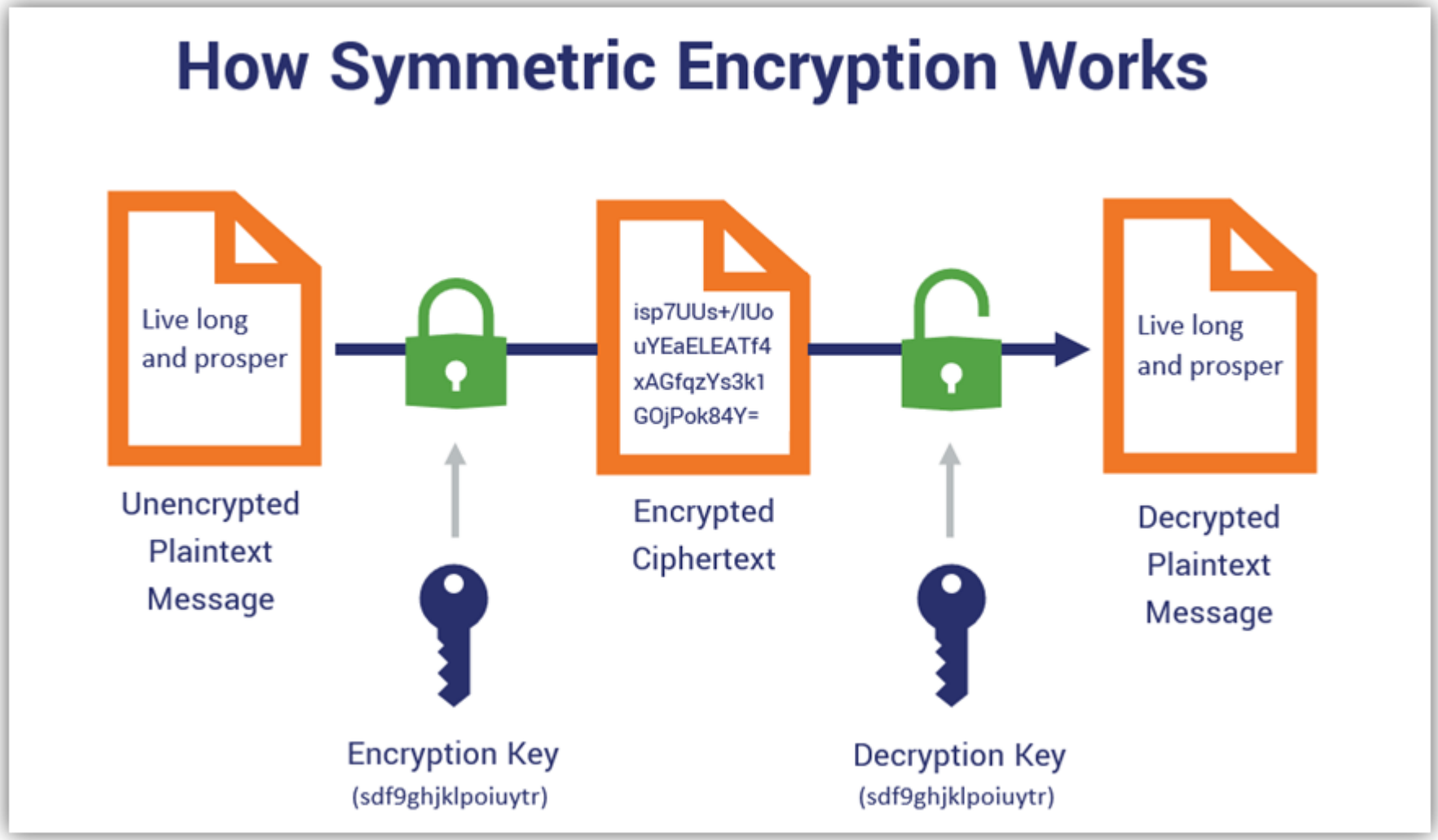
- The sender encrypts data from the upper layer using the encryption keys
- The receiver decrypts the data using the same key
- Encrypted frame body is meaningless to a third party
- Management and control frames are usually not encrypted





WLAN Encryption (ctd)

- WLAN encryption is symmetric because the keys used for encryption and decryption are identical
- The keys used are PTK (Pairwise Transient Key) and GTK (Group Temporal Key a.k.a. Groupwise Transient Key)



RSN Authentication

- After authentication, another frame exchange, called the 4-Way Handshake, must also occur to create dynamic encryption keys for both radios
  - This should not be confused with the 4-frame exchange used in DCF
- A pairwise master key (PMK) is generated as a by-product of either PSK, SAE, or 802.1X/EAP authentication
- The PMK is used to seed the 4-Way Handshake that generates the unique dynamic encryption keys

RSN Authentication methods

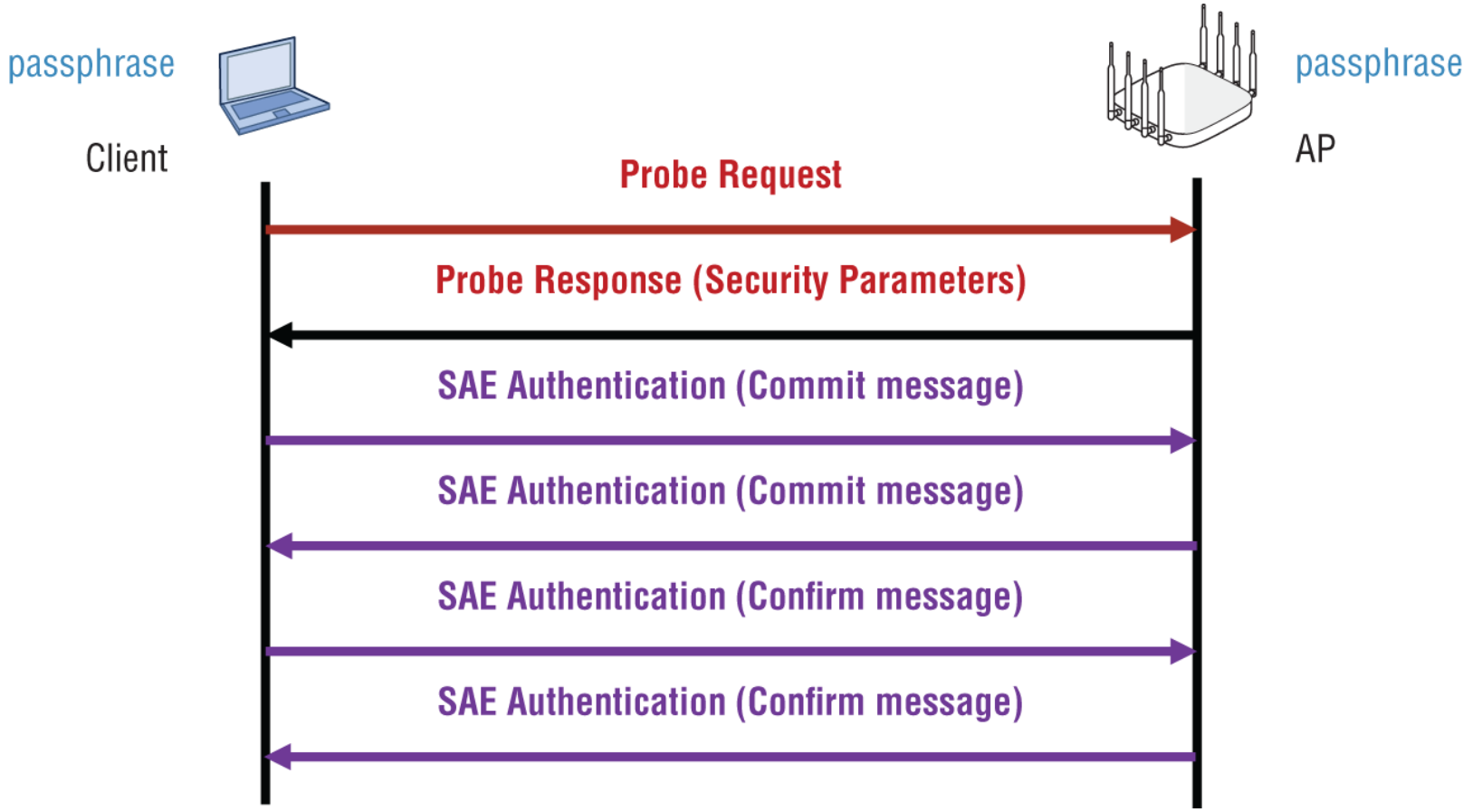
Method	Authentication	Encryption	Cypher	Key Generation
WPA Personal	PSK	Temporal Key Integrity Protocol (TKIP)	ARC4	Dynamic
WPA Enterprise	802.1X / EAP	TKIP	ARC4	Dynamic
WPA2 Personal	PSK	CCMP (mandatory) TKIP ((optional)	AES (mandatory) or ARC4 (optional)	Dynamic
WPA3 Personal	Simultaneous Authentication of Equals (SAE)	CCMP (mandatory) or TKIP ((optional)	AES (mandatory) or ARC4 (optional)	Dynamic
WPA3 Enterprise	802.1X / EAP	CCMP (mandatory)	AES (mandatory)	Dynamic

PSK Authentication

- Used in Small Office and Home networks
- Wi-Fi Protected Access - WPA/WPA2-Personal allows a simple ASCII character string, called a passphrase of length 8 to 63 characters to be used
- The passphrase, together with the SSID is used to generate a 256-bit PSK
  - Example:
    Passphrase: Carolina
    SSID: cwna
    PSK: 7516b6d5169ca633ece6aa43e0ca9d5c0afa08268ab9fde47c38a627546b71c5
- The passphrase used on the client and the AP must match
- A pairwise master key (PMK) is generated as a by-product of either PSK, SAE, or 802.1X/EAP authentication
- The PMK is used to seed the 4-Way Handshake that generates the unique dynamic encryption keys employed by any two 802.11 radios

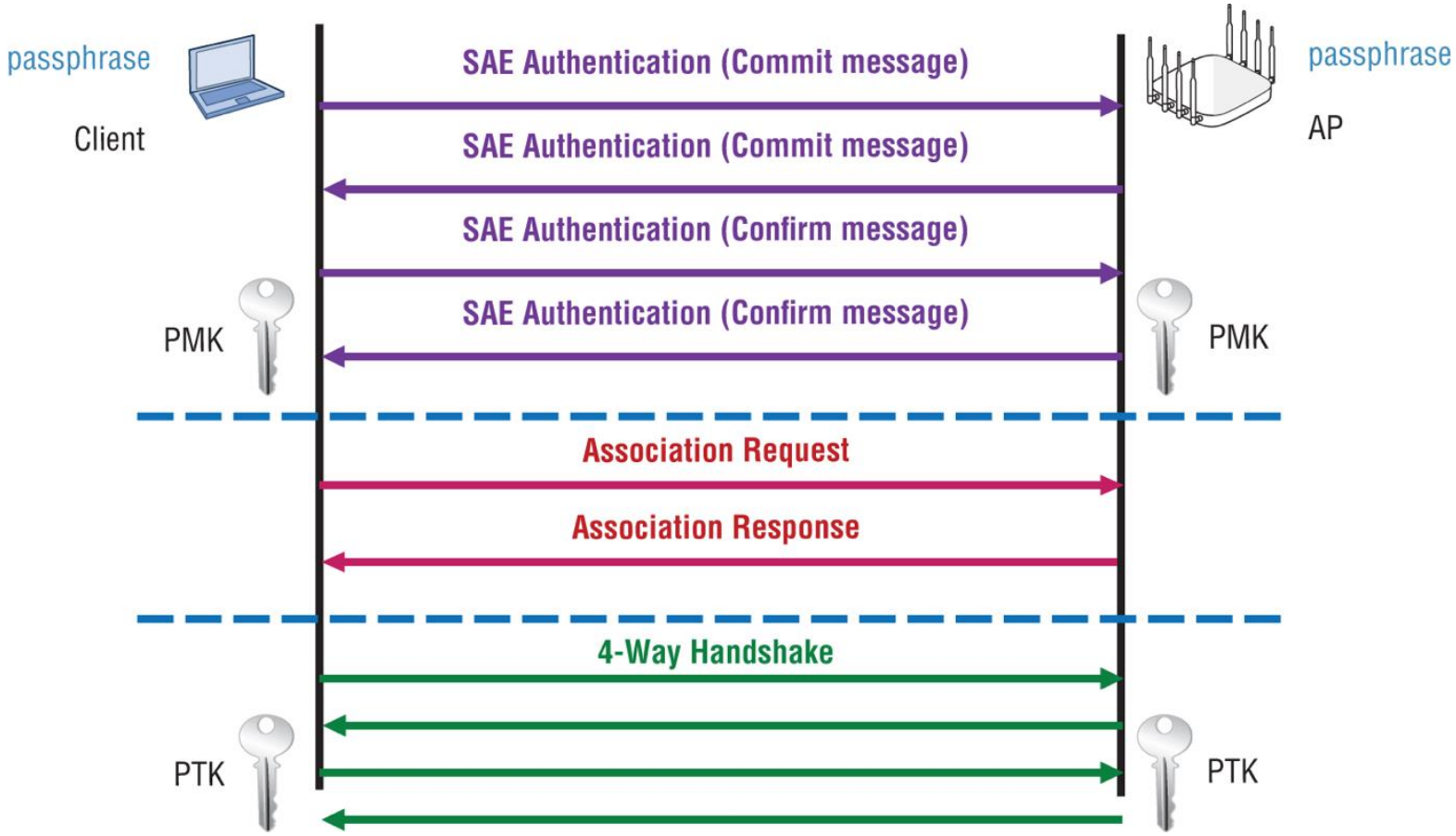
SAE Authentication

- SAE protocol exchange protects the passphrase from brute-force dictionary attacks
- The passphrase is never sent between 802.11 stations during the SAE exchange
- Consists of a commitment message exchange and a confirmation message exchange



SAE Authentication (ctd)

- Once the SAE exchanges are complete, a unique pairwise master key (PMK) is derived and installed on both the AP and the client station
- The PMK is the seeding material for the 4-Way Handshake that is used to create create a pairwise transient key (PTK)

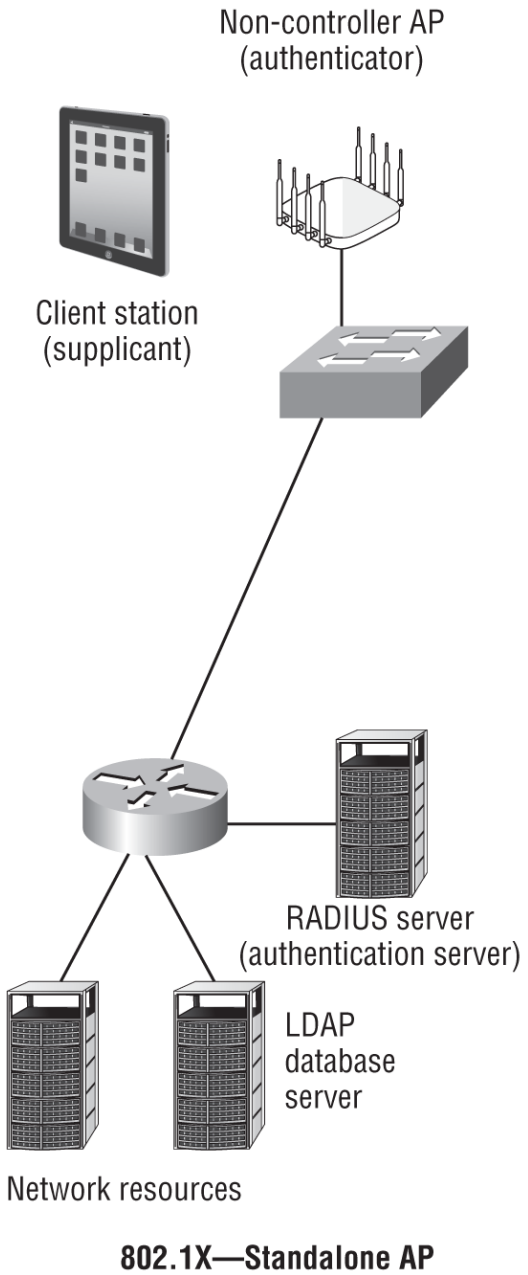


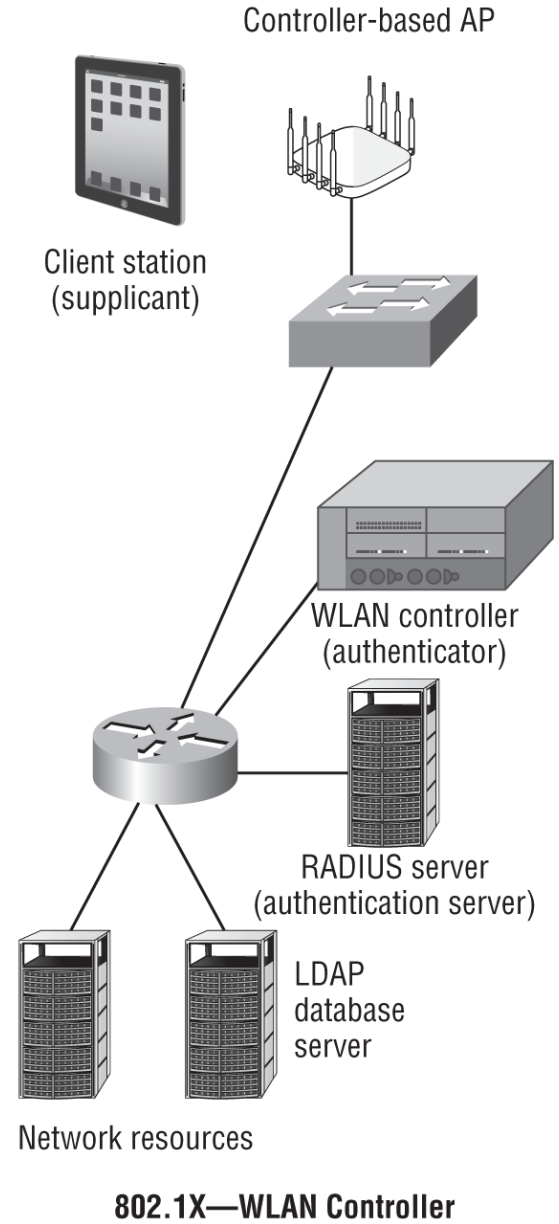
802.1X/EAP Authentication

- Used in enterprise networks
- The 802.1X authorization framework consists of three main components
  - Supplicant: device that is seeking access to the network
  - Authenticator: device that blocks or permits the traffic from supplicant
  - Authentication server: Validates the credentials provided by the supplicant and notifies the authenticator that the supplicant has been authorized
- These components work together to ensure that only properly validated users and devices are authorized to access network resources
- A layer 2 authentication protocol called Extensible Authentication Protocol (EAP) is used within the 802.1X framework to validate users at layer 2

802.1X/EAP Authentication

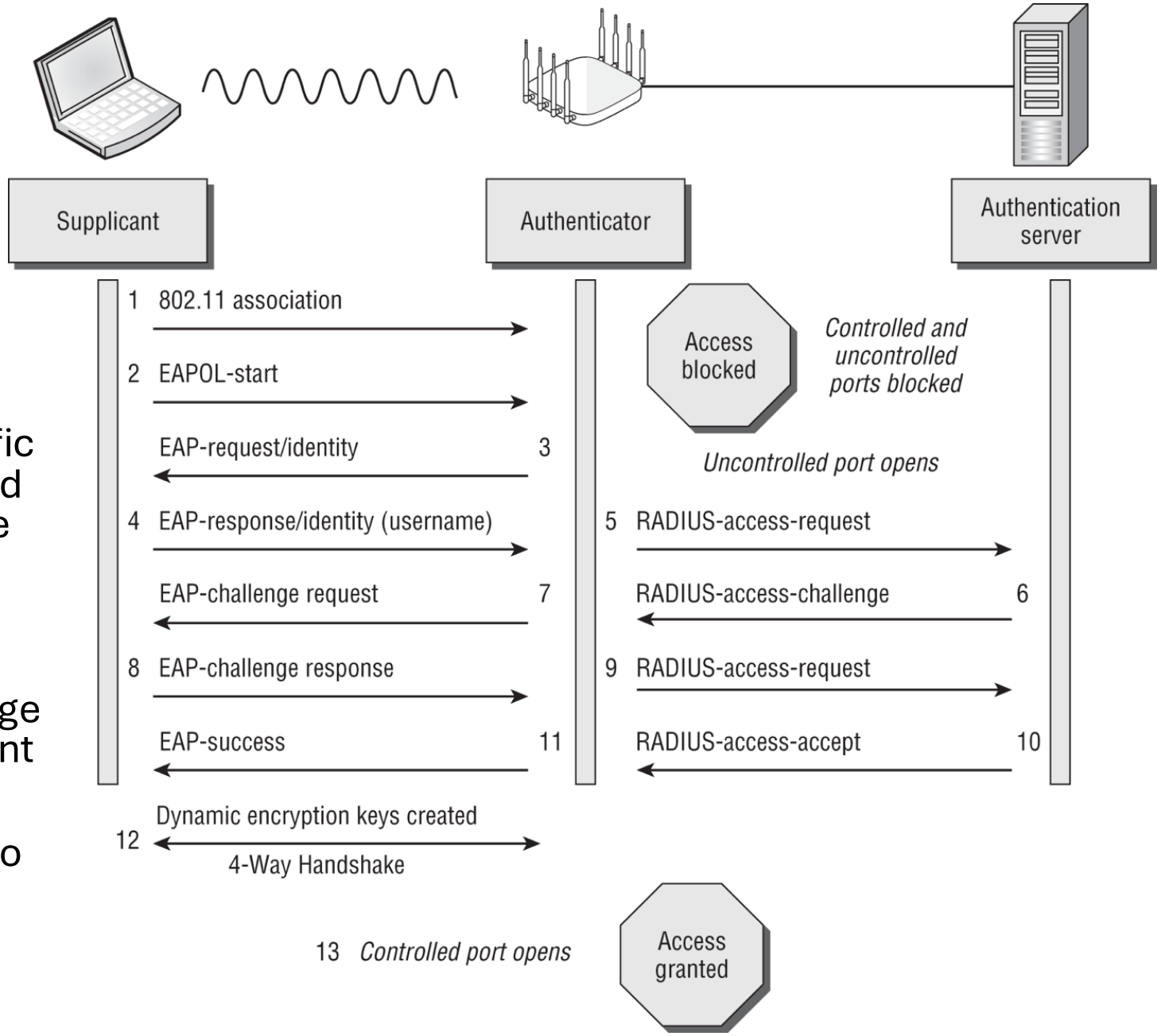
- Authenticator is usually an AP or a WLAN Controller
- A RADIUS server normally functions as the authentication server
- The authentication server may use a username / password database, such as an LDAP database, to authenticate the supplicant credentials





802.1X/EAP Authentication (2)

- Extensible Authentication Protocol (EAP) is a layer 2 protocol used to provide user authentication
- The authenticator allows the EAP traffic to pass through its virtual uncontrolled port, but blocks all other traffic via the virtual controlled port
- After the authentication server has verified the credentials of the supplicant, the server sends a message to the authenticator that the supplicant has been authenticated
- The authenticator is then authorized to open the virtual controlled port and allow all other traffic to pass through



4-Way Handshake

- In Robust Security Network (RSN), two stations (STAs) must establish a procedure to authenticate and associate with each other as well as create dynamic encryption keys
- This process is known as the 4-Way Handshake
- Note that the PTK is derived independently by both entities
- GTK is derived by the authenticator, and sent to the supplicant encrypted using PTK

